POSITIVITY OF ROTATING BLACK HOLE INDICES ON K3

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Abstract. We prove Sen's conjecture that single-centred rotating black holes in type IIB string theory on $K3 \times S^1$ have a positive index at fixed angular momentum, for every charge of discriminant at most 1000 and for all charges whose D1–D5 product is 1, 2, 3, 5, 7, 11 or 13.

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1. Introduction

Strominger and Vafa explained the entropy of a five-dimensional supersymmetric black hole by counting the bound states of D1-branes, D5-branes and momentum on $K3 \times S^1$ [24], and Breckenridge, Myers, Peet and Vafa extended the count to black holes that rotate [5]. The number that is computed at weak coupling is an index: states are weighted by a sign, so that the result is protected when the coupling grows and the black hole forms. Dijkgraaf, Verlinde and Verlinde found the generating function of these indices for type II string theory on $K3 \times T^2$: it is the reciprocal of the Igusa cusp form Φ_{10} of genus two [13]. Through the relation between four- and five-dimensional black holes found by Gaiotto, Strominger and Yin [15], the same function governs the rotating black holes on $K3 \times S^1$.

The integers obtained in this way count all bound states, not only the black hole. Some of them are bound states of two black holes held apart by electromagnetic forces, and some are modes that live outside the horizon. Sen showed that once both are removed, what remains is the index of the horizon itself, and he argued that this index must be positive: a supersymmetric single-centred black hole in four dimensions is spherically symmetric, so its microstates carry zero angular momentum and contribute to the index with the same sign [22, 11]. For the rotating black holes of five dimensions he defined the index at fixed angular momentum, which removes the multiplicities coming from rotations of the horizon, and conjectured that it too is positive [23]. Bringmann and Murthy turned these predictions into precise statements about Fourier coefficients and proved them when the D1–D5 product $m = Q_1 Q_5$ equals 1 or 2 [6].[1]

This paper proves the conjecture at fixed angular momentum for every charge of discriminant at most 1000, and for all charges when $m \in \{1, 2, 3, 5, 7, 11, 13\}$.

The indices. Throughout, τ lies in the upper half-plane $\mathbb{H}$, $z \in \mathbb{C}$, $y = \operatorname{Im} \tau$, $\mathbf{e}(x) = e^{2\pi i x}$, $q = \mathbf{e}(\tau)$ and $\zeta = \mathbf{e}(z)$. Let $\Delta = q \prod_{n \geq 1} (1 - q^n)^{24}$, and let $p_{24}(k)$ be the number of partitions of k into parts of 24 colours, so that $\Delta^{-1} = \sum_{k \geq 0} p_{24}(k) q^{k-1}$. Let

$$A(\tau, z) = (\zeta - 2 + \zeta^{-1}) \prod_{n \geq 1} \frac{(1 - q^n \zeta)^2 (1 - q^n \zeta^{-1})^2}{(1 - q^n)^4}$$

[1] Bringmann and Murthy give two proofs for $m = 2$, one by the circle method and one by explicit identities between theta functions, Eisenstein series and class numbers [6, §§4–6]. They state the fixed angular momentum conjecture in [6, §2.3], attributing it to [23].

be the weak Jacobi form $\phi_{-2,1}$ of weight -2 and index 1, and let $\phi_{0,1}$ be the weak Jacobi form of weight 0 and index 1 normalised by $\phi_{0,1}(\tau,0)=12$; the elliptic genus of a K3 surface is $2\phi_{0,1}$ [14, §9], [12]. Define weak Jacobi forms χ_N of weight 0 and index N by

$$\sum_{N\geq 0}\chi_N(\tau,z)\,p^N=\exp\Big(\sum_{k\geq 1}p^k\,(2\phi_{0,1})\big|V_k\,(\tau,z)\Big), \tag{1}$$

where V_k is the Hecke operator of weight 0 recalled in Section 4; by the theorem of Dijkgraaf, Moore, Verlinde and Verlinde, χ_N is the elliptic genus of the symmetric product $\mathrm{Sym}^N(K3)$ [12]. For $m\geq -1$ put

$$\psi_m=\frac{\chi_{m+1}}{\Delta\,A}. \tag{2}$$

These are the coefficients of the expansion $1/\Phi_{10}=\sum_{m\geq -1}\psi_m(\tau,z)\,\mathbf{e}(m\sigma)$ in the third variable of the Siegel upper half-space [6, §3.1]. The function ψ_m has double poles at the points $z\in\mathbb{Z}\tau+\mathbb{Z}$, and its Fourier coefficients depend on where it is expanded; this dependence is the wall crossing of two-centred black holes. Following Dabholkar, Murthy and Zagier as presented in [6, §3.1], one removes the poles with the Appell–Lerch sum

$$A_{1,m}(\tau,z)=\sum_{s\in\mathbb{Z}}\frac{q^{ms^2+s}\zeta^{2ms+1}}{(1-q^s\zeta)^2}, \tag{3}$$

and puts

$$\psi_m^F=\psi_m-\frac{p_{24}(m+1)}{\Delta}\,A_{1,m}=\sum_{n\geq -1}\sum_{l\in\mathbb{Z}}c_m^F(n,l)\,q^n\zeta^l. \tag{4}$$

The function ψ_m^F is holomorphic in z (Lemma 4.2), so its coefficients $c_m^F(n,l)$ are well defined. They are integers with $c_m^F(n,l)=c_m^F(n,-l)$, and $c_m^F(n,l)$ depends only on $4mn-l^2$ and on l modulo $2m$.

A four-dimensional charge is described by three integers (n,l,m), the T-duality invariants $Q^2/2$, $Q\cdot P$ and $P^2/2$ of the electric and magnetic charge vectors, and its discriminant is $D=4mn-l^2$. A single-centred black hole exists only for $D>0$, and its horizon area is then $4\pi\sqrt{D}$ in Planck units, so that its entropy is $\pi\sqrt{D}$. To the charge belongs the positive definite binary quadratic form $nX^2+lXY+mY^2$. Every such form is equivalent under $\mathrm{GL}_2(\mathbb{Z})$ to exactly one form $aX^2+bXY+cY^2$ with $0\leq b\leq c\leq a$ (Lemma 3.2), and we call (a,b,c) the reduced form of (n,l,m). The *single-centred index* is

$$d^*(n,l,m)=\begin{cases}(-1)^{b+1}\,c_c^F(a,b), & D>0,\\ 0, & D\leq 0,\end{cases} \tag{5}$$

and Section 2 explains why this is the index of the horizon. For the five-dimensional black hole with momentum n, angular momentum $J_{3L}=J/2$ and D1–D5 product $m=Q_1Q_5$, the index at fixed angular momentum is [23], [6, §2.3, §3.1]

$$d(n,J,m)=d^*(n,J,m)-d^*(n,J+2,m),\qquad J\geq 0. \tag{6}$$

The conjecture of Sen is that $d(n,J,m)>0$ whenever $4mn-J^2>0$; it contains the four-dimensional statement $d^*>0$ as the special case in which the second term vanishes.

Results. Let $\mathcal{M} = \{1, 2, 3, 5, 7, 11, 13\}$.

Theorem A (Charges of small discriminant)**.** *For every charge with $0 < 4mn - l^2 \le 1000$ the single-centred index $d^*(n, l, m)$ is positive, and for every five-dimensional charge with $0 < 4mn - J^2 \le 1000$ the index $d(n, J, m)$ at fixed angular momentum is positive.*

Theorem B (Fixed angular momentum)**.** *Let $m \in \mathcal{M}$. Then $d(n, J, m) > 0$ for all integers n and $J \ge 0$ with $4mn - J^2 > 0$.*

Corollary C. *If the reduced form (a, b, c) of a charge (n, l, m) with $4mn - l^2 > 0$ has $c \in \mathcal{M}$, then $d^*(n, l, m) > 0$.*

The proof of Theorem B needs an exact description of the mock modular nature of ψ_m^F. Let $H(N)$ be the Hurwitz class number of positive definite binary quadratic forms of discriminant $-N$, counted with weight $1/2$ and $1/3$ for forms equivalent to multiples of $X^2 + Y^2$ and $X^2 + XY + Y^2$, with $H(0) = -1/12$ and $H(N) = 0$ for $N < 0$ or $N \equiv 1, 2 \pmod 4$. Let $\mathcal{H}(\tau, z) = \sum_{n,l} H(4n - l^2) q^n \zeta^l$, and let $\mathcal{H}|V_m$ be its image under the Hecke operator of weight 2, whose coefficients are

$$h_m(n, l) = \sum_{d|(m,n,l)} d\, H\Big(\frac{4mn - l^2}{d^2}\Big). \tag{7}$$

Put $\lambda_1 = 648$ and $\lambda_p = p_{24}(p + 1)$ for a prime p.

Theorem D (The class number term)**.** *For $m = 1$ and for every prime m, the function*

$$\phi_m = \Delta\, \psi_m^F + \lambda_m\, \mathcal{H}|V_m$$

is a weak Jacobi form of weight 2 and index m. For $m \in \mathcal{M}$ it is the polynomial in $E_4, E_6, A, \phi_{0,1}$ listed in Appendix A; in particular $\phi_1 = 3E_4A$ and $\phi_2 = \frac{1}{3}(22E_4A\phi_{0,1} - 10E_6A^2)$.

For $m = 1, 2$ Theorem D is the pair of identities [6, (3.6), (3.7)]. For a prime m the constant λ_m comes out of the proof: it is the unique multiple of $\mathcal{H}|V_m$ whose nonholomorphic completion cancels that of $A_{1,m}$. The value $\lambda_1 = 2p_{24}(2)$, rather than $p_{24}(2)$, reflects the fact that the operator V_p doubles the shadow of $\mathcal{H}$ while V_1 does not.

What the proof has to overcome. Three features make the problem harder than its four-dimensional analogue in $\mathcal{N} = 8$ string theory, where the index is a finite combination of coefficients of a single weak Jacobi form.

First, the coefficients $c_m^F(n, l)$ themselves are not positive in the required sense. For $m = 5$ the charge $(n, J) = (10, 14)$ has discriminant 4, and the coefficient of ψ_5^F at this discriminant gives $(-1)^{14+1} c_5^F(10, 14) = -2\,023\,536$; the reduced form of $(10, 14, 5)$ is $(1, 0, 1)$, and $d^*(10, 14, 5) = -c_1^F(1, 0) = 50\,064$. The passage to the reduced form in (5) is therefore essential, and it has the consequence that the two terms of (6) can be coefficients of two different functions ψ_c^F.

Second, ψ_m^F is not modular: it is the sum of a weakly holomorphic Jacobi form and a multiple of $\mathcal{H}|V_m/\Delta$. The main term of the asymptotic formula comes from the first, but the class numbers contribute an error that is only polynomially smaller in the range that matters for $m = 1, 2, 3$.

Third, the positivity of $d(n, J, m)$ is a statement about a difference of two exponentially large numbers. When J is small the two discriminants D and $D - 4(J + 1)$ are close, and the main terms differ by a factor $1 - O(D^{-1/2})$. The error terms have to be smaller than this relative gap, uniformly, and they are controlled only beyond an explicit discriminant D_m^* that ranges from 258 to 1030.

Plan. Section 2 recalls where the formula (6) comes from. Section 3 proves the facts about binary quadratic forms that the reduction needs, in particular a lemma on successive minima that controls the second charge $(n, J+2, m)$. Section 4 defines the Jacobi forms and proves that ψ_m^F is holomorphic, elliptic and even, and that its most polar coefficient is $m+2$. Section 5 proves Theorem D from Zwegers' completion of Appell–Lerch sums and Zagier's completion of the class number generating function. Section 6 gives the exact formula for the coefficients of ϕ_m/Δ, Section 7 proves the inequalities for Bessel functions that are needed, and Section 8 proves the tail estimate: beyond D_m^* the index grows faster than its error. Sections 9 and 10 prove Theorems B and A and Corollary C, and Section 11 discusses what lies beyond. Appendix A lists ϕ_m and the polar coefficients, Appendix B describes the exact computations, and Appendix C proves the elementary bounds for $H(N)$ and $p_{24}(k)$.

K3 surfaces, their involutions and the Calabi–Yau threefolds fibred by them are studied in [3, 4], and the heterotic and M-theory compactifications to which type II strings on K3 are dual in [20].

2. Black holes and their indices

This section explains the formulae (5) and (6). Nothing in the proofs of Sections 3 to 10 depends on it, and the reader interested only in the mathematics can take (4), (5) and (6) as definitions.

2.1. **Four dimensions.** Type IIB string theory on $K3 \times S^1 \times \widetilde{S}^1$ has $\mathcal{N}=4$ supersymmetry in four dimensions. Its charges form a vector (Q, P) of electric and magnetic charges in two copies of an even lattice of signature $(6, 22)$, and the duality group $\mathrm{SL}_2(\mathbb{Z}) \times O(6,22;\mathbb{Z})$ acts on them. The T-duality invariants

$$n = \tfrac{1}{2}Q^2, \qquad l = Q \cdot P, \qquad m = \tfrac{1}{2}P^2$$

are integers, and the S-duality group $\mathrm{SL}_2(\mathbb{Z})$ acts on the triple (n, l, m) exactly as $\mathrm{SL}_2(\mathbb{Z})$ acts on the coefficients of the binary quadratic form $nX^2 + lXY + mY^2$ by linear substitutions of (X, Y) [21]. A quarter-BPS black hole with these charges exists when the discriminant $D = 4mn - l^2$ is positive, and its entropy is then $\pi\sqrt{D}$.

The index computed by the microscopic theory is a helicity trace, the quantity that remains constant as the string coupling is increased until a black hole forms. For charges with torsion one, the indices of all states of charge (n, l, m) are, up to the sign $(-1)^{l+1}$, the Fourier coefficients of $1/\Phi_{10}$ in the variables conjugate to n, l and m [13, 21]. Because $1/\Phi_{10}$ has poles, these coefficients depend on the contour of integration, and the contour is determined by the moduli at infinity. The jumps across the poles are the indices of bound states of two black holes, which exist on one side of a wall in moduli space and not on the other [21].

The single-centred black hole is isolated by choosing the moduli at their attractor values, where no two-centred bound state exists. Dabholkar, Murthy and Zagier showed that for a charge with $0 \le l \le m \le n$ the index at the attractor point is the coefficient of ψ_m^F; Bringmann and Murthy record this as [6, §3.1]

$$d^*(n, l, m) = (-1)^{l+1} c_m^F(n, l) \qquad (0 \le l \le m \le n,\ 4mn - l^2 > 0).$$

The single-centred index does not depend on the moduli at infinity, so it is invariant under S-duality, which moves the moduli but not the attractor values belonging to a given charge. Every charge with $D > 0$ is S-dual, up to the reflection $l \mapsto -l$, to a unique charge with $0 \le l \le m \le n$ (Lemma 3.2); the reflection does not change the index because $c_m^F(n, -l) = c_m^F(n, l)$. This gives (5) for $D > 0$. A charge with $D \le 0$ carries no single-centred black hole, and its single-centred index is zero.

2.2. **Five dimensions.** In type IIB string theory on $K3 \times S^1$, a system of Q_5 D5-branes wrapped on $K3 \times S^1$ and Q_1 D1-branes wrapped on S^1, carrying n units of momentum along S^1, forms a supersymmetric black hole in five dimensions whose horizon rotates with angular momentum $J_{3L} = J/2$ in one SU(2) factor of the rotation group; this is the black hole of Breckenridge, Myers, Peet and Vafa [5]. Gaiotto, Strominger and Yin observed that placing it at the centre of a Taub–NUT space relates it to the four-dimensional black hole with invariants (n, J, m), $m = Q_1 Q_5$ [15]. After the modes that live outside the horizon are removed [2, 10], the index of the five-dimensional black hole at fixed $J_{3L} = J/2$, summed over all values of $\vec{J}_L^2$, is $d^*(n, J, m)$ [6, §2.3, §3.1].

The states of the black hole form multiplets of the rotation group $\mathrm{SU}(2)_L$. A multiplet of spin j contributes to the index at every value $J_{3L} \in \{-j, -j+1, \dots, j\}$, so the index at $J_{3L} = J/2$ counts the multiplets of spin $j \geq J/2$, and the difference (6) counts those of spin exactly $J/2$. These are the states with fixed J_{3L} and fixed $\vec{J}_L^2$. Sen introduced this index and argued that for a black hole of given angular momentum its horizon states contribute with a single sign, so that $d(n, J, m)$ must be positive whenever the black hole exists [23].[2] Since $4mn - (J+2)^2 < 4mn - J^2$, the second term of (6) vanishes when $4mn - (J+2)^2 \leq 0$, and then $d(n, J, m) = d^*(n, J, m)$.

3. Binary quadratic forms

We write $[a, b, c]$ for the form $aX^2 + bXY + cY^2$ and $Q(v)$ for its value at $v = (x, y) \in \mathbb{Z}^2$. All forms in this section are positive definite, with discriminant $D = 4ac - b^2 > 0$. The group $\mathrm{GL}_2(\mathbb{Z})$ acts by $Q \mapsto Q \circ g$; this changes the coefficients but not D or the set of values. A form is *reduced* if $0 \leq b \leq c \leq a$.[3]

Lemma 3.1. *Let $Q = [a, b, c]$ be reduced. Then:*

(i) *$Q(x, y) \geq a$ for all $(x, y) \in \mathbb{Z}^2$ with $x \neq 0$;*
(ii) *$Q(v) \geq c$ for all $v \neq 0$, with equality at $v = \pm(0, 1)$;*
(iii) *the only primitive vectors v with $Q(v) < a$ are $\pm(0, 1)$;*
(iv) *$D \geq 3c^2$ and $D \leq 4ac$.*

Proof. (i) Let $x \neq 0$. If $|y| \leq |x|$, then $b|xy| \leq bx^2$, so $Q(x, y) \geq (a - b)x^2 + cy^2$. When $y = 0$ this equals $ax^2 \geq a$; when $y \neq 0$ it is at least $(a - b) + c \geq a$ because $c \geq b$ and $a \geq b$. If $|y| > |x|$, then $b|xy| \leq by^2$, so $Q(x, y) \geq ax^2 + (c - b)y^2 \geq a$.

(ii) For $x \neq 0$ this follows from (i) and $a \geq c$; for $x = 0$ we have $Q(0, y) = cy^2 \geq c$ with equality exactly when $y = \pm 1$.

(iii) By (i) a vector with $Q(v) < a$ has $x = 0$, and the primitive vectors with $x = 0$ are $\pm(0, 1)$.

(iv) $D = 4ac - b^2 \geq 4c^2 - c^2 = 3c^2$ because $a \geq c \geq b$, and $D \leq 4ac$ because $b^2 \geq 0$. □

For a positive definite form Q let $\lambda_1(Q)$ be its smallest nonzero value and $\lambda_2(Q)$ the smallest number t such that two linearly independent vectors v, w have $Q(v) \leq t$ and $Q(w) \leq t$. Both are invariant under $\mathrm{GL}_2(\mathbb{Z})$.

Lemma 3.2. *Every positive definite binary quadratic form is $\mathrm{GL}_2(\mathbb{Z})$-equivalent to exactly one reduced form $[a, b, c]$, and $c = \lambda_1(Q)$, $a = \lambda_2(Q)$, $b = \sqrt{4ac - D}$.*

[2] The sign conventions of (5) and (6) are those of [6]: the factor $(-1)^{l+1}$ removes the sign carried by the fermion zero modes of the broken supersymmetries, and with it the prediction is that both d^* and d are positive.

[3] The usual reduced forms for $\mathrm{SL}_2(\mathbb{Z})$ are those with $|b| \leq c \leq a$ and $b \geq 0$ when $|b| = c$ or $c = a$ [9, §2.2]; here the variable with the smaller value is Y, as in the charge (n, l, m), and the reflection $b \mapsto -b$ is allowed.

Proof. For a reduced form, Lemma 3.1(ii) gives $\lambda_1 = c$. Two linearly independent vectors cannot both have first coordinate 0, so one of them has value at least a by Lemma 3.1(i), while $(1,0)$ and $(0,1)$ have values a and $c \le a$; hence $\lambda_2 = a$. Since D, λ_1 and λ_2 are invariants and $b \ge 0$, two equivalent reduced forms coincide.

For existence, let v_1 be a vector at which Q takes its smallest nonzero value; it is primitive. Among the vectors v_2 for which (v_2, v_1) is a basis of $\mathbb{Z}^2$, choose one with $Q(v_2)$ smallest; such vectors exist because v_1 is primitive, and Q takes only finitely many values below any bound. In the basis (v_2, v_1) the form is $[a, b', c]$ with $a = Q(v_2)$, $c = Q(v_1)$ and $a \ge c$. The vectors $v_2 \pm v_1$ also complete v_1 to a basis, so $a \pm b' + c = Q(v_2 \pm v_1) \ge a$, that is, $|b'| \le c$. Replacing v_1 by $-v_1$ if necessary, which is a substitution of determinant -1, we obtain $b = |b'|$ and a reduced form. □

The proof is the classical reduction of Lagrange and Gauss; an efficient algorithm alternates the swap $[a,b,c] \mapsto [c,-b,a]$ with the translation $[a,b,c] \mapsto [a - bk + ck^2,\ b - 2ck,\ c]$, $k = \lfloor (b+c)/(2c) \rfloor$, until $|b| \le c \le a$ [9, §2.2]. This is how the reduced forms are computed in Appendix B.

The next lemma controls the reduced form of a charge in terms of the number it represents at $(0,1)$. It is the reason why the proof of Theorem B needs the coefficients of ψ_c^F only for $c \le m$.

Lemma 3.3. *Let Q be a positive definite form that takes the value m at a primitive vector, and let $[a,b,c]$ be its reduced form. Then $c = m$ or $a \le m$. In particular, if $D > 4m^2$, then $c = m$.*

Proof. Transport the primitive vector v_0 with $Q(v_0) = m$ to the coordinates in which Q is reduced. By Lemma 3.1(iii), either $v_0 = \pm(0,1)$, and then $m = Q(v_0) = c$, or $m = Q(v_0) \ge a$. In the second case $c \le a \le m$ and Lemma 3.1(iv) gives $D \le 4ac \le 4m^2$. □

Both forms that enter (6), $[n, J, m]$ and $[n, J+2, m]$, take the value m at the primitive vector $(0,1)$, so Lemma 3.3 applies to both.

4. Jacobi forms and the finite part

4.1. **Jacobi forms.** Let $k \in \mathbb{Z}$ and $m \ge 1$. Following Eichler and Zagier [14] and Zwegers [27, Ch. 3], consider functions ϕ on $\mathbb{C} \times \mathbb{H}$, real-analytic or meromorphic, with the elliptic and modular transformation laws

$$\text{(E)} \qquad \phi(\tau, z + \lambda\tau + \mu) = \mathbf{e}\big(-m(\lambda^2\tau + 2\lambda z)\big)\,\phi(\tau, z) \qquad (\lambda, \mu \in \mathbb{Z}),$$

$$\text{(M)} \qquad \phi\Big(\frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d}\Big) = (c\tau + d)^k\, \mathbf{e}\Big(\frac{mcz^2}{c\tau + d}\Big)\phi(\tau, z) \qquad \Big(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})\Big).$$

A *weak Jacobi form* of weight k and index m is a holomorphic function on $\mathbb{H} \times \mathbb{C}$ satisfying (E) and (M) with a Fourier expansion $\sum_{n \ge 0} \sum_l c(n,l) q^n \zeta^l$. Its coefficients satisfy $c(n,l) = 0$ unless $4mn - l^2 \ge -m^2$ [14, §9]. A *weakly holomorphic* Jacobi form is a holomorphic function satisfying (E) and (M) whose expansion has $n \ge n_0$ for some n_0, such as ϕ/Δ for a weak Jacobi form ϕ.

The Eisenstein series $E_4 = 1 + 240 \sum \sigma_3(n) q^n$ and $E_6 = 1 - 504 \sum \sigma_5(n) q^n$ are Jacobi forms of index 0. The form $A = \phi_{-2,1}$ of the introduction equals $-\vartheta_1(\tau, z)^2 / \eta(\tau)^6$, and

$$A = (\zeta - 2 + \zeta^{-1}) + q\,(-2\zeta^2 + 8\zeta - 12 + 8\zeta^{-1} - 2\zeta^{-2}) + O(q^2),$$
$$2\phi_{0,1} = (2\zeta + 20 + 2\zeta^{-1}) + q\,(20\zeta^2 - 128\zeta + 216 - 128\zeta^{-1} + 20\zeta^{-2}) + O(q^2).$$

The coefficient of $q^n\zeta^l$ in $2\phi_{0,1}$ depends only on $4n-l^2$; we write it $C(4n-l^2)$, so that $C(-1)=2$, $C(0)=20$, $C(3)=-128$, $C(4)=216$, and $C(D)=0$ for $D<-1$. Eichler and Zagier proved that weak Jacobi forms of even weight form the polynomial ring

$$\mathbb{C}[E_4,E_6][A,\phi_{0,1}], \tag{8}$$

so that the weak Jacobi forms of weight k and index m have the basis of monomials $E_4^\alpha E_6^\beta A^i \phi_{0,1}^j$ with $i+j=m$ and $4\alpha+6\beta-2i=k$ [14, Thm. 9.3].

4.2. **Theta decomposition.** For $l\in\mathbb{Z}/2m\mathbb{Z}$ let

$$\vartheta_{m,l}(\tau,z)=\sum_{r\equiv l\ (2m)} q^{r^2/4m}\zeta^r.$$

If ϕ is holomorphic in z and satisfies (E), then [27, Thm. 3.1], [14, §5]

$$\phi(\tau,z)=\sum_{l \bmod 2m} h_l(\tau)\,\vartheta_{m,l}(\tau,z),\qquad h_l(\tau)=\mathbf{e}\Big(-\frac{l^2\tau}{4m}\Big)\int_p^{p+1}\phi(\tau,z)\,\mathbf{e}(-lz)\,dz, \tag{9}$$

for any $p\in\mathbb{C}$. Writing $\phi=\sum_r g_r(\tau)\zeta^r$, the law (E) with $\lambda=1$ gives $g_{r+2m}(\tau)=q^{((r+2m)^2-r^2)/4m}g_r(\tau)$. Consequently, if $\phi=\sum c(n,l)q^n\zeta^l$, the coefficient $c(n,l)$ depends only on $4mn-l^2$ and on l modulo $2m$, and $h_l(\tau)=\sum_n c(n,l)q^{n-l^2/4m}$. If ϕ also satisfies (M), then

$$h_l(\tau+1)=\mathbf{e}\Big(-\frac{l^2}{4m}\Big)h_l(\tau),\qquad h_l\Big(-\frac{1}{\tau}\Big)=\frac{\tau^k}{\sqrt{-i\tau}}\,\frac{1}{\sqrt{2m}}\sum_{j \bmod 2m}\mathbf{e}\Big(\frac{lj}{2m}\Big)h_j(\tau) \tag{10}$$

[27, (3.3), (3.4)], so $(h_l)_l$ is a vector-valued modular form of weight $k-\frac12$.

4.3. **Hecke operators.** For a function ϕ satisfying (E) and (M) with weight k and index m, and for $N\ge1$, Eichler and Zagier define

$$(\phi|_kV_N)(\tau,z)=N^{k-1}\sum_{\substack{ad=N\\ b \bmod d}} d^{-k}\,\phi\Big(\frac{a\tau+b}{d},az\Big).$$

The function $\phi|_kV_N$ satisfies (E) and (M) with weight k and index mN; the proof of [14, Thm. 4.2] uses only these two laws, so it applies to real-analytic functions as well. If $\phi=\sum c(n,l)q^n\zeta^l$ and $m=1$, then

$$\phi|_kV_N=\sum_{n,l}\Big(\sum_{d|(n,l,N)} d^{k-1}c\Big(\frac{nN}{d^2},\frac{l}{d}\Big)\Big)q^n\zeta^l \tag{11}$$

[14, Thm. 4.2]. We write V_N for $|_0V_N$ when it acts on $2\phi_{0,1}$, as in (1), and for $|_2V_N$ when it acts on $\mathcal{H}$, as in (7).

4.4. **The functions χ_N.** Taking the logarithm of a product shows that (1) is equivalent to

$$\sum_{N\ge0}\chi_N\,p^N=\prod_{N\ge1}\prod_{n\ge0}\prod_{l\in\mathbb{Z}}(1-p^Nq^n\zeta^l)^{-C(4nN-l^2)}. \tag{12}$$

Indeed, the logarithm of the right side is $\sum_{N,n,l}C(4nN-l^2)\sum_{r\ge1}p^{Nr}q^{nr}\zeta^{lr}/r$, and its coefficient of p^k is $\sum_{r|k}r^{-1}\sum_{n,l}C(4nk/r-l^2)q^{nr}\zeta^{lr}$, which is $(2\phi_{0,1})|_0V_k$ by (11). Since the exponents in (12) are integers, χ_N has integer coefficients. Since $(2\phi_{0,1})|V_k$ is a weak Jacobi form of weight 0 and index k, χ_N is a weak Jacobi form of weight 0 and index N.

Lemma 4.1. (i) *$\chi_N(\tau,0)=p_{24}(N)$ for all τ.*

(ii) *The coefficient of q^0 in χ_N is a Laurent polynomial $\chi_N^{(0)}(\zeta)$ of degree N, invariant under $\zeta \mapsto \zeta^{-1}$, whose coefficient of ζ^N is $N+1$.*

Proof. (i) Since $2\phi_{0,1}(\tau,0) = 24$, the definition of V_k gives

$$(2\phi_{0,1})|V_k(\tau,0) = \frac{1}{k}\sum_{ad=k}\sum_{b \bmod d} 24 = \frac{24\,\sigma(k)}{k}.$$

Hence

$$\sum_N \chi_N(\tau,0)p^N = \exp\Big(24\sum_{k\geq 1}\frac{\sigma(k)}{k}p^k\Big) = \exp\Big(24\sum_{d,e\geq 1}\frac{p^{de}}{e}\Big) = \prod_{d\geq 1}(1-p^d)^{-24}.$$

(ii) In (12) only the factors with $n=0$ contribute to q^0, and $C(-l^2)$ is 20 for $l=0$, 2 for $l=\pm 1$ and 0 otherwise. So

$$\sum_N \chi_N^{(0)}(\zeta)p^N = \prod_{e\geq 1}(1-p^e\zeta)^{-2}(1-p^e)^{-20}(1-p^e\zeta^{-1})^{-2}.$$

A monomial in the expansion of this product that contributes to p^N has ζ-degree at most N, with equality only when every factor used is $(1-p\zeta)^{-2}$; the coefficient of $(p\zeta)^N$ in $(1-p\zeta)^{-2}$ is $N+1$. The symmetry under $\zeta \mapsto \zeta^{-1}$ is visible in the product. □

4.5. **The functions ψ_m and ψ_m^F.** By (2), ψ_m is a meromorphic function satisfying (E) and (M) with weight $0-(-2)-12 = -10$ and index $(m+1)-1 = m$. The function A vanishes exactly at $z \in \mathbb{Z}\tau+\mathbb{Z}$, to second order: from the product, $A(\tau,z) = (\zeta^{1/2}-\zeta^{-1/2})^2(1+O(z^2)) = (2\pi i z)^2(1+O(z^2))$ near $z=0$, where the $O(z^2)$ term is even in z because A is. So ψ_m has double poles at the lattice points and no other poles.

In the strip $0 < \operatorname{Im} z < \operatorname{Im}\tau$, that is $|q| < |\zeta| < 1$, one has

$$\frac{1}{A} = \frac{\zeta}{(1-\zeta)^2}\prod_{n\geq 1}\frac{(1-q^n)^4}{(1-q^n\zeta)^2(1-q^n\zeta^{-1})^2},$$

and each factor expands in nonnegative powers of q, $q\zeta$ and $q\zeta^{-1}$, with $\zeta/(1-\zeta)^2 = \sum_{k\geq 1} k\zeta^k$. Hence $\Delta\psi_m = \chi_{m+1}/A$ has an expansion $\sum_{n\geq 0}\sum_l c(n,l)q^n\zeta^l$ in the strip. The sum (3) converges absolutely and locally uniformly away from its poles, because the terms with $s \to +\infty$ are $O(|q|^{ms^2})$ and those with $s \to -\infty$ are $O(|q|^{ms^2-s}|\zeta|^{2ms-1})$ after the identity $(1-x)^{-2} = x^{-2}(1-x^{-1})^{-2}$. In the strip, the terms with $s \geq 0$ expand as

$$\frac{q^{ms^2+s}\zeta^{2ms+1}}{(1-q^s\zeta)^2} = \sum_{k\geq 0}(k+1)\,q^{ms^2+s+sk}\zeta^{2ms+1+k},$$

and those with $s<0$, for which $|q^s\zeta| > 1$, as

$$\frac{q^{ms^2+s}\zeta^{2ms+1}}{(1-q^s\zeta)^2} = \sum_{k\geq 0}(k+1)\,q^{ms^2-s-sk}\zeta^{2ms-1-k}.$$

In both cases the exponent n of q is nonnegative and $4mn-l^2 = -(k+1)^2 < 0$, as a direct computation shows. These are the expansions used in Appendix B.

Lemma 4.2. *The function ψ_m^F is holomorphic on $\mathbb{H}\times\mathbb{C}$, satisfies (E) with index m, and is even in z. Its expansion (4) has integer coefficients, $c_m^F(n,l) = 0$ for $n < -1$, and $c_m^F(n,l)$ depends only on $4mn-l^2$ and on l modulo $2m$, with $c_m^F(n,l) = c_m^F(n,-l)$.*

Proof. Ellipticity. Replacing z by $z+1$ does not change (3). Replacing z by $z+\tau$ multiplies ζ by q, and since $ms^2+s+2ms+1 = m(s+1)^2+(s+1)-m$, the term with index s becomes $q^{-m}\zeta^{-2m}$ times the term with index $s+1$. So $A_{1,m}$ satisfies (E) with index m, and so does ψ_m.

Parity. The functions χ_{m+1} and A are even in z, hence so is ψ_m. In (3), replacing z by $-z$ and using $(1-q^s\zeta^{-1})^{-2} = q^{-2s}\zeta^2(1-q^{-s}\zeta)^{-2}$ turns the term of index s into the term of index $-s$, so $A_{1,m}$ is even as well.

Poles. The poles of $A_{1,m}$ lie where $q^s\zeta = 1$ for some s, that is at $z \in \mathbb{Z}\tau+\mathbb{Z}$, so both functions in (4) have their poles in the lattice. By (E) it suffices to examine the points $z = -s\tau$, $s \in \mathbb{Z}$. Write $z = -s\tau+\varepsilon$ and $x = 2\pi i\varepsilon$. The term of index s of (3) is then

$$q^{ms^2+s}\, q^{-2ms^2-s} e^{2msx} e^x \,(1-e^x)^{-2} = q^{-ms^2} e^{2msx}\, \frac{e^x}{(1-e^x)^2},$$

while the other terms are holomorphic at $\varepsilon = 0$. On the other hand (E) gives $\psi_m(\tau, -s\tau+\varepsilon) = q^{-ms^2}e^{2msx}\psi_m(\tau,\varepsilon)$. Hence, near $\varepsilon = 0$,

$$\psi_m^F(\tau, -s\tau+\varepsilon) = q^{-ms^2}e^{2msx}\Big(\psi_m(\tau,\varepsilon) - \frac{p_{24}(m+1)}{\Delta(\tau)}\frac{e^x}{(1-e^x)^2}\Big) + (\text{holomorphic}).$$

The two functions in the bracket are even in ε: the first because ψ_m is even, the second because $e^x/(1-e^x)^2 = (e^{x/2}-e^{-x/2})^{-2}$. Their principal parts are $\chi_{m+1}(\tau,0)\Delta^{-1}x^{-2}$ and $p_{24}(m+1)\Delta^{-1}x^{-2}$, which agree by Lemma 4.1(i). An even meromorphic function with no x^{-2} term and a pole of order at most two has no pole, since it cannot have an x^{-1} term. So the bracket is holomorphic, and so is ψ_m^F.

Coefficients. Since ψ_m^F is holomorphic in z, its expansion in powers of ζ is the same in every strip, and we may compute it in $|q| < |\zeta| < 1$, where $\Delta\psi_m$ and $A_{1,m}$ have the integral expansions above with $n \geq 0$; multiplying by $\Delta^{-1} = \sum_k p_{24}(k)q^{k-1}$ gives integer coefficients with $n \geq -1$. The remaining statements follow from the theta decomposition (9) and from parity. □

Figure 1 shows the poles and the strip in which the expansions are computed.

Lemma 4.3. *Let $m = 1$ or let m be prime. If $l, l' \in \{0, 1, \dots, m\}$ and $l^2 \equiv l'^2 \pmod{4m}$, then $l = l'$. Consequently there is a function F_m on the integers $D \equiv -l^2 \pmod{4m}$ such that*

$$(-1)^{l+1}c_m^F(n,l) = F_m(4mn-l^2) \qquad \text{for all } n, l,$$

and if the reduced form of a charge (n, l, m') with $D > 0$ is $[a, b, m]$, then $d^*(n, l, m') = F_m(D)$.

Proof. For $m = 1$ the squares 0 and 1 are distinct modulo 4. Let m be prime and $l^2 \equiv l'^2 \pmod{4m}$. Then m divides $(l-l')(l+l')$. If $m \mid l-l'$, then $l = l'$ or $\{l, l'\} = \{0, m\}$, and the latter is impossible because $m^2 \not\equiv 0 \pmod{4m}$ for a prime m. If $m \mid l+l'$, then $l+l' \in \{0, m, 2m\}$. The cases 0 and $2m$ give $l = l'$. In the case $l+l' = m$ we have $l'^2-l^2 = m(m-2l)$, so $4 \mid m-2l$; this is impossible for odd m, and for $m = 2$ the only possibility with $l \neq l'$ is $\{l, l'\} = \{0, 2\}$, where $l'^2 - l^2 = \pm 4 \not\equiv 0 \pmod 8$.

By Lemma 4.2, $c_m^F(n,l)$ depends only on $D = 4mn-l^2$ and on the class of $\pm l$ modulo $2m$, which has a unique representative in $\{0, \dots, m\}$; by the first part, this representative is determined by D. The sign $(-1)^{l+1}$ is determined by D too, because $D \equiv 0 \pmod 4$ when l is even and $D \equiv 3 \pmod 4$ when l is odd. The last statement is the definition (5). □

For composite m the value of $c_m^F(n,l)$ is not determined by $4mn-l^2$; for instance, for $m = 4$ the residues $l = 0$ and $l = 4$ both give $D \equiv 0 \pmod{16}$, and at $D = 16$ one finds $c_4^F(1,0) = -185\,738\,352$ but $c_4^F(2,4) = -32\,861\,184$.

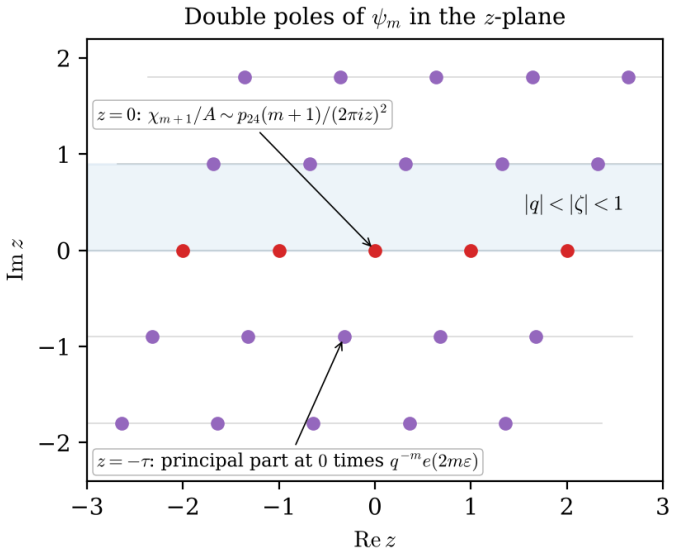


FIGURE 1. The double poles of ψ_m at $z \in \mathbb{Z}\tau + \mathbb{Z}$, here for $\tau = 0.32 + 0.9i$, and the strip $|q| < |\zeta| < 1$ in which $\Delta\psi_m$ and $A_{1,m}$ are expanded. Near $z = 0$ the principal part of ψ_m is $p_{24}(m+1)\Delta^{-1}(2\pi i z)^{-2}$; near $z = -s\tau$ it is the same expression multiplied by $q^{-ms^2}e^{2ms\cdot 2\pi i\varepsilon}$, which produces a simple pole as well. The Appell–Lerch sum $p_{24}(m+1)A_{1,m}/\Delta$ has exactly these principal parts (Lemma 4.2).

Lemma 4.4. *The coefficient of q^{-1} in ψ_m^F is the Laurent polynomial*

$$\big(\chi_{m+1}^{(0)}(\zeta) - p_{24}(m+1)\big)\frac{\zeta}{(1-\zeta)^2},$$

which has degree m and leading coefficient $m+2$. In particular $c_m^F(-1,\pm m) = m+2$ and $c_m^F(-1,l) = 0$ for $|l| > m$.

Proof. In the strip, the coefficient of q^{-1} in $\psi_m = \Delta^{-1}\chi_{m+1}/A$ is the coefficient of q^0 in χ_{m+1}/A, which is $\chi_{m+1}^{(0)}(\zeta)\,\zeta(1-\zeta)^{-2}$. In the expansion of $A_{1,m}$ above, the exponent n vanishes only for $s = 0$, where the terms are $(k+1)\zeta^{k+1}$; so the coefficient of q^0 in $A_{1,m}$ is $\zeta(1-\zeta)^{-2}$. This proves the formula, since $\chi_{m+1}^{(0)}(1) = \chi_{m+1}(\tau,0)|_{q^0} = p_{24}(m+1)$ by Lemma 4.1(i). The numerator vanishes at $\zeta = 1$, and so does its derivative because it is invariant under $\zeta \mapsto \zeta^{-1}$; hence it is divisible by $(1-\zeta)^2$ and the quotient is a Laurent polynomial. By Lemma 4.1(ii) the numerator has degree $m+1$ with leading coefficient $m+2$, the factor ζ raises the degree by one, and division by $(1-\zeta)^2 = \zeta^2(1-\zeta^{-1})^2$ lowers it by two. □

5. The completion and the class number term

This section proves Theorem D. The idea is that $A_{1,m}$ and $\mathcal{H}|V_m$ are both holomorphic parts of real-analytic functions that transform like Jacobi forms of weight 2 and index m, and that their nonholomorphic parts are proportional. We use the error function in the normalisation of Zwegers,

$$E(x) = 2\int_0^x e^{-\pi t^2}\,dt = \operatorname{erf}(\sqrt{\pi}\,x),$$

and the function

$$I(x)=\int_1^\infty u^{-3/2}e^{-xu}\,du=2e^{-x}-2\sqrt{\pi x}\,\operatorname{erfc}\sqrt{x}\qquad(x\ge 0),\tag{13}$$

where the second expression follows by integrating by parts once and substituting $t=xu$ in $\int_1^\infty u^{-1/2}e^{-xu}du=x^{-1/2}\int_x^\infty t^{-1/2}e^{-t}dt=\sqrt{\pi/x}\,\operatorname{erfc}\sqrt{x}$. For $m\ge 1$ and $l\in\mathbb{Z}/2m\mathbb{Z}$ put

$$S_{m,l}(\tau)=y^{-1/2}\sum_{\lambda\equiv l\ (2m)} I\Big(\frac{\pi\lambda^2 y}{m}\Big)q^{-\lambda^2/4m}.\tag{14}$$

Since $0<I(x)\le 2e^{-x}$ and $|q^{-\lambda^2/4m}|=e^{\pi\lambda^2 y/2m}$, the terms are bounded by $2y^{-1/2}e^{-\pi\lambda^2 y/2m}$ and the series converges absolutely and locally uniformly in τ.

5.1. **Zwegers' building block.** For $u\in\mathbb{C}$ Zwegers defines [27, Def. 3.2]

$$f_u(z;\tau)=\sum_{\lambda\in\mathbb{Z}}\frac{q^{m\lambda^2}\zeta^{2m\lambda}}{1-q^\lambda\zeta\,\mathbf{e}(-u)},$$

a meromorphic function of z with simple poles at $u+\mathbb{Z}\tau+\mathbb{Z}$, and the real-analytic modification [27, Def. 3.4]

$$\tilde f_u(z;\tau)=f_u(z;\tau)-\frac12\sum_{l\bmod 2m}R_{m,l}(u;\tau)\,\vartheta_{m,l}(\tau,z),$$

$$R_{m,l}(u;\tau)=\sum_{\lambda\equiv l\ (2m)}\Big\{\operatorname{sgn}\big(\lambda+\tfrac12\big)-E\Big(\big(\lambda+2m\tfrac{\operatorname{Im}u}{y}\big)\sqrt{y/m}\Big)\Big\}\,\mathbf{e}\Big(-\frac{\lambda^2\tau}{4m}-\lambda u\Big).$$

For u in a compact set and $|\lambda|$ large the bracket is bounded by $2\exp(-\pi(\lambda+2m\operatorname{Im}u/y)^2y/m)$, because $1-E(x)=\operatorname{erfc}(\sqrt{\pi}x)\le e^{-\pi x^2}$ for $x\ge 0$, while the exponential factor has modulus $\exp(\pi\lambda^2y/2m+2\pi\lambda\operatorname{Im}u)$; so the series for $R_{m,l}$ and its termwise derivatives in u converge locally uniformly. Zwegers proves [27, Prop. 3.5] that $\tilde f_u$ satisfies (E) in z with index m and that

$$\tilde f_{u/(c\tau+d)}\Big(\frac{z}{c\tau+d};\frac{a\tau+b}{c\tau+d}\Big)=(c\tau+d)\,\mathbf{e}\Big(\frac{mc(z^2-u^2)}{c\tau+d}\Big)\tilde f_u(z;\tau)\tag{15}$$

for all $\left(\begin{smallmatrix}a&b\\c&d\end{smallmatrix}\right)\in\mathrm{SL}_2(\mathbb{Z})$. Write $\partial_u=\frac12(\partial_{\operatorname{Re}u}-i\partial_{\operatorname{Im}u})$, so that $\partial_u u=1$, $\partial_u\bar u=0$ and $\partial_u\operatorname{Im}u=1/(2i)$.

Lemma 5.1. *For $z\notin\mathbb{Z}\tau+\mathbb{Z}$,*

$$\partial_u f_u(z;\tau)\big|_{u=0}=-2\pi i\,A_{1,m}(\tau,z),\qquad \partial_u R_{m,l}(u;\tau)\big|_{u=0}=i\sqrt{m}\,S_{m,l}(\tau).$$

Proof. With $w=q^\lambda\zeta$ we have $\partial_u(1-w\,\mathbf{e}(-u))^{-1}=-2\pi i\,w\,\mathbf{e}(-u)(1-w\,\mathbf{e}(-u))^{-2}$, which is $-2\pi i\,w(1-w)^{-2}$ at $u=0$. Multiplying by $q^{m\lambda^2}\zeta^{2m\lambda}$ and summing gives the first formula, since $q^{m\lambda^2}\zeta^{2m\lambda}\cdot q^\lambda\zeta=q^{m\lambda^2+\lambda}\zeta^{2m\lambda+1}$.

For the second, differentiate the term of index λ at $u=0$ and put $X=\pi\lambda^2y/m$. Since $E'(x)=2e^{-\pi x^2}$, the derivative of the bracket is

$$-2e^{-\pi\lambda^2y/m}\cdot\frac{2m}{y}\cdot\frac{1}{2i}\cdot\sqrt{\frac{y}{m}}=2i\sqrt{\frac{m}{y}}\,e^{-X},$$

and the derivative of the exponential factor is $-2\pi i\lambda$ times that factor. At $u=0$ the bracket equals $\operatorname{sgn}(\lambda+\frac12)-E(\lambda\sqrt{y/m})$, which is $\operatorname{erfc}\sqrt{X}$ for $\lambda\ge 0$ and $-\operatorname{erfc}\sqrt{X}$ for $\lambda<0$; times λ it equals $|\lambda|\operatorname{erfc}\sqrt{X}$ in both cases. Hence the derivative of the term is

$$i\Big(2\sqrt{\frac{m}{y}}\,e^{-X}-2\pi|\lambda|\operatorname{erfc}\sqrt{X}\Big)q^{-\lambda^2/4m}=i\sqrt{\frac{m}{y}}\Big(2e^{-X}-2\sqrt{\pi X}\operatorname{erfc}\sqrt{X}\Big)q^{-\lambda^2/4m},$$

because $\sqrt{y/m}\cdot\pi|\lambda|=\sqrt{\pi X}$. By (13) this is $i\sqrt{m}\,y^{-1/2}I(X)q^{-\lambda^2/4m}$, and summing over $\lambda\equiv l \pmod{2m}$ gives $i\sqrt{m}\,S_{m,l}$. □

Lemma 5.2. *The function*

$$\widehat{A}_{1,m}(\tau,z)=A_{1,m}(\tau,z)+\frac{\sqrt{m}}{4\pi}\sum_{l \bmod 2m}S_{m,l}(\tau)\,\vartheta_{m,l}(\tau,z)$$

is real-analytic in τ, meromorphic in z with poles only at $\mathbb{Z}\tau+\mathbb{Z}$, and satisfies (E) *and* (M) *with weight* 2 *and index m.*

Proof. By Lemma 5.1 and the definition of $\tilde f_u$,

$$-\frac{1}{2\pi i}\,\partial_u\tilde f_u\big|_{u=0}=A_{1,m}+\frac{1}{4\pi i}\sum_l i\sqrt{m}\,S_{m,l}\vartheta_{m,l}=\widehat{A}_{1,m}.$$

The law (E) for $\tilde f_u$ in z does not involve u, so it passes to $\partial_u\tilde f_u$. For (M), apply ∂_u to (15) and put $u=0$. The left side is a function of $u/(c\tau+d)$, which is holomorphic in u; by the chain rule its derivative is $(c\tau+d)^{-1}$ times the derivative of $\tilde f$ in its first argument, evaluated at $u=0$. On the right side the factor $\mathbf{e}(-mcu^2/(c\tau+d))$ has derivative zero at $u=0$. Hence

$$(\partial_u\tilde f)\Big(\frac{z}{c\tau+d};\frac{a\tau+b}{c\tau+d}\Big)=(c\tau+d)^2\,\mathbf{e}\Big(\frac{mcz^2}{c\tau+d}\Big)(\partial_u\tilde f)(z;\tau)$$

at $u=0$, which is (M) with $k=2$. The poles of f_u at $z=u+\mathbb{Z}\tau+\mathbb{Z}$ become double poles at $\mathbb{Z}\tau+\mathbb{Z}$ after differentiation, and the theta functions are entire. □

5.2. **Zagier's completion of the class numbers.** Let $F(\tau)=\sum_{N\ge0}H(N)q^N$. Zagier proved that

$$\widehat{F}(\tau)=F(\tau)+\frac{1}{16\pi\sqrt{y}}\sum_{f\in\mathbb{Z}}I(4\pi f^2y)\,q^{-f^2} \tag{16}$$

satisfies $\widehat{F}(\gamma\tau)=j(\gamma,\tau)^3\,\widehat{F}(\tau)$ for every $\gamma\in\Gamma_0(4)$, where $j(\gamma,\tau)=\theta(\gamma\tau)/\theta(\tau)$ and $\theta(\tau)=\sum_{n\in\mathbb{Z}}q^{n^2}$; that is, $\widehat F$ transforms like θ^3, a modular form of weight $3/2$ on $\Gamma_0(4)$ [26], [16, Ch. 2, Thm. 2]. In [16] the nonholomorphic term is written $y^{-1/2}\sum_f\beta(4\pi f^2y)q^{-f^2}$ with $\beta(x)=\frac{1}{16\pi}\int_1^\infty u^{-3/2}e^{-xu}du=I(x)/16\pi$. The quotient $j(\gamma,\tau)$ is holomorphic, because θ has no zeros in $\mathbb{H}$ by the product $\theta(\tau)=\prod_{n\ge1}(1-q^{2n})(1+q^{2n-1})^2$.

We also need the theta function in two variables, $\theta(\sigma,w)=\sum_{n\in\mathbb{Z}}\mathbf{e}(n^2\sigma+nw)$, so that $\theta(\sigma)=\theta(\sigma,0)$. It satisfies $\theta(\sigma,w+1)=\theta(\sigma,w)$, $\theta(\sigma,w+\frac12)=\theta(\sigma+\frac12,w)$ (since $\mathbf{e}(n/2)=\mathbf{e}(n^2/2)$), and $\theta(\sigma,w+2\sigma)=\mathbf{e}(-\sigma-w)\,\theta(\sigma,w)$ (replace n by $n-1$). Moreover, with the principal branch of the square root,

$$\theta\Big(\frac{\sigma}{4\sigma+1},\frac{w}{4\sigma+1}\Big)=(4\sigma+1)^{1/2}\,\mathbf{e}\Big(\frac{w^2}{4\sigma+1}\Big)\theta(\sigma,w). \tag{17}$$

To see this, put $\Theta(t,z)=\sum_n\exp(\pi in^2t+2\pi inz)$, so that $\theta(\sigma,w)=\Theta(2\sigma,w)$. Poisson summation applied to the Gaussian gives Jacobi's formula $\Theta(-1/t,z/t)=(-it)^{1/2}\exp(\pi iz^2/t)\,\Theta(t,z)$

for $t \in \mathbb{H}$, and clearly $\Theta(t-2,z) = \Theta(t,z)$ and $\Theta(t,-z) = \Theta(t,z)$. Let $t \in \mathbb{H}$ and $t_2 = -1/t - 2 \in \mathbb{H}$, so that $-1/t_2 = t/(2t+1)$. Applying Jacobi's formula at t_2 with z/t, then the periodicity, then Jacobi's formula at t,

$$\Theta\Big(\frac{t}{2t+1}, \frac{-z}{2t+1}\Big) = (-it_2)^{1/2}(-it)^{1/2}\exp\Big(\pi i z^2\Big(\frac{1}{t^2 t_2} + \frac{1}{t}\Big)\Big)\Theta(t,z).$$

Here $t^2 t_2 = -t(2t+1)$, so the exponent is $2\pi i z^2/(2t+1)$; and $-it$, $-it_2$ lie in the right half-plane, so the product of their principal square roots is the principal square root of $(-it_2)(-it) = 2t+1$. Evenness in z and $t = 2\sigma$, $z = w$ give (17). At $w = 0$ it says $j(\gamma_1, \sigma) = (4\sigma+1)^{1/2}$ for $\gamma_1 = \left(\begin{smallmatrix} 1 & 0 \\ 4 & 1 \end{smallmatrix}\right)$.

For $\mu \in \mathbb{Z}/2\mathbb{Z}$ put $H_\mu(\tau) = \sum_{N \equiv -\mu^2 \ (4)} H(N) q^{N/4}$, so that $\mathcal{H} = \sum_\mu H_\mu \vartheta_{1,\mu}$ by (9), and

$$\widehat{H}_\mu = H_\mu + \frac{1}{8\pi} S_{1,\mu}, \qquad \widehat{\mathcal{H}} = \sum_{\mu \bmod 2} \widehat{H}_\mu\, \vartheta_{1,\mu} = \mathcal{H} + \frac{1}{8\pi} \sum_{\mu \bmod 2} S_{1,\mu}\vartheta_{1,\mu}.$$

Lemma 5.3. *The function $\widehat{\mathcal{H}}$ is real-analytic, entire in z, and satisfies* (E) *and* (M) *with weight* 2 *and index* 1.

Proof. From (14), $S_{1,\mu}(4\tau) = (4y)^{-1/2} \sum_{\lambda \equiv \mu \ (2)} I(4\pi\lambda^2 y) q^{-\lambda^2}$, so

$$\widehat{H}_0(4\tau) + \widehat{H}_1(4\tau) = \sum_N H(N) q^N + \frac{1}{16\pi\sqrt{y}} \sum_{\lambda \in \mathbb{Z}} I(4\pi\lambda^2 y) q^{-\lambda^2} = \widehat{F}(\tau).$$

The exponents of q in $\widehat{H}_\mu$ are $N/4$ with $N \equiv -\mu^2 \pmod 4$, and $-\lambda^2/4$ with $\lambda \equiv \mu \pmod 2$, so $\widehat{H}_\mu(\tau+1) = \mathbf{e}(-\mu^2/4)\widehat{H}_\mu(\tau)$. Hence $\widehat{H}_0(\tau) + \widehat{H}_1(\tau) = \widehat{F}(\sigma)$ and $\widehat{H}_0(\tau) - \widehat{H}_1(\tau) = \widehat{H}_0(\tau+2) + \widehat{H}_1(\tau+2) = \widehat{F}(\sigma + \frac{1}{2})$, where $\sigma = \tau/4$. On the other side $\vartheta_{1,0} + \vartheta_{1,1} = \sum_r \mathbf{e}(r^2\tau/4 + rz) = \theta(\sigma, z)$ and $\vartheta_{1,0} - \vartheta_{1,1} = \theta(\sigma, z + \frac{1}{2}) = \theta(\sigma + \frac{1}{2}, z)$. Therefore

$$\widehat{\mathcal{H}}(\tau, z) = \tfrac{1}{2}\big(\widehat{F}(\sigma)\,\theta(\sigma, z) + \widehat{F}(\sigma')\,\theta(\sigma', z)\big), \qquad \sigma' = \sigma + \tfrac{1}{2}.$$

The law (E) holds for every function $\sum_\mu h_\mu(\tau)\vartheta_{1,\mu}(\tau, z)$, since each $\vartheta_{1,\mu}$ satisfies it. For (M), the matrices $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $U = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ generate $\mathrm{SL}_2(\mathbb{Z})$, because $T^{-1}UT^{-1} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and the factor in (M) is a cocycle, so it suffices to check (M) for T and U. For T it holds because $\widehat{H}_\mu(\tau+1)\vartheta_{1,\mu}(\tau+1, z) = \widehat{H}_\mu(\tau)\vartheta_{1,\mu}(\tau, z)$. For U, which sends (τ, z) to $(\tau/(\tau+1), z/(\tau+1))$, note that σ goes to $\gamma_1\sigma$ and $\tau + 1 = 4\sigma + 1$. By (17) and the law of $\widehat{F}$,

$$\widehat{F}(\gamma_1\sigma)\,\theta\Big(\gamma_1\sigma, \frac{z}{4\sigma+1}\Big) = (4\sigma+1)^2\,\mathbf{e}\Big(\frac{z^2}{4\sigma+1}\Big)\widehat{F}(\sigma)\,\theta(\sigma, z).$$

For the second term, $\gamma_1\sigma + \frac{1}{2} = (3\sigma + \frac{1}{2})/(4\sigma+1) = \gamma_2\sigma'$ with $\gamma_2 = \left(\begin{smallmatrix} 3 & -1 \\ 4 & -1 \end{smallmatrix}\right) \in \Gamma_0(4)$, whose lower row gives $4\sigma' - 1 = 4\sigma + 1$. Writing $\frac{1}{2} = (2\sigma + \frac{1}{2})/(4\sigma+1)$ and applying (17) with $w = z + 2\sigma + \frac{1}{2}$, then the quasi-periodicity,

$$\theta\Big(\gamma_2\sigma', \frac{z}{4\sigma+1}\Big) = \theta\Big(\gamma_1\sigma, \frac{z + 2\sigma + \frac{1}{2}}{4\sigma+1}\Big) = (4\sigma+1)^{1/2}\,\mathbf{e}\Big(\frac{(z + 2\sigma + \frac{1}{2})^2}{4\sigma+1} - \sigma - z - \frac{1}{2}\Big)\theta(\sigma', z),$$

and the exponent equals $z^2/(4\sigma+1) - \frac{1}{4}$, as one checks by expanding $(z + 2\sigma + \frac{1}{2})^2 - (\sigma + z + \frac{1}{2})(4\sigma+1) = z^2 - \sigma - \frac{1}{4}$. At $z = 0$ this gives $j(\gamma_2, \sigma') = -i(4\sigma+1)^{1/2}$. Hence

$$\widehat{F}(\gamma_2\sigma')\,\theta\Big(\gamma_2\sigma', \frac{z}{4\sigma+1}\Big) = (-i)^4(4\sigma+1)^2\,\mathbf{e}\Big(\frac{z^2}{4\sigma+1}\Big)\widehat{F}(\sigma')\,\theta(\sigma', z).$$

Adding the two identities gives $\widehat{\mathcal{H}}(U(\tau, z)) = (\tau+1)^2\mathbf{e}(z^2/(\tau+1))\widehat{\mathcal{H}}(\tau, z)$, which is (M) for U with $k = 2$ and $m = 1$. Real-analyticity is clear from the definition, and $\widehat{\mathcal{H}}$ is entire in z because the theta functions are. □

Lemma 5.4. *Let p be a prime. Then*

$$\widehat{\mathcal{H}}|_2 V_p = \mathcal{H}|V_p + \frac{\sqrt{p}}{4\pi} \sum_{l \bmod 2p} S_{p,l}\, \vartheta_{p,l},$$

where $\mathcal{H}|V_p$ has the coefficients (7).

Proof. The operator V_p is linear and is applied termwise to absolutely convergent series. Its action on $\mathcal{H}$ is given by (11) with $k = 2$, which gives (7). It remains to compute $G|_2 V_p$ for

$$G(\tau, z) = \frac{1}{8\pi} \sum_{\mu} S_{1,\mu} \vartheta_{1,\mu} = \frac{1}{8\pi\sqrt{y}} \sum_{\lambda \in \mathbb{Z}} \sum_{r \equiv \lambda\,(2)} I(\pi\lambda^2 y)\, \mathbf{e}\Big(\frac{(r^2-\lambda^2)\tau}{4}\Big) \zeta^r.$$

By definition $G|_2 V_p = p \sum_{ad=p} \sum_{b \bmod d} d^{-2} G\big(\frac{a\tau+b}{d}, az\big)$. The term with $a = p$, $d = 1$ is $pG(p\tau, pz)$; replacing y by py and writing $\lambda' = p\lambda$, $r' = pr$, it equals

$$\frac{\sqrt{p}}{8\pi\sqrt{y}} \sum_{p|\lambda'} \sum_{r' \equiv \lambda'\,(2p)} I\Big(\frac{\pi\lambda'^2 y}{p}\Big) \mathbf{e}\Big(\frac{(r'^2-\lambda'^2)\tau}{4p}\Big) \zeta^{r'}.$$

The terms with $a = 1$, $d = p$ give $p^{-1} \sum_{b \bmod p} G((\tau+b)/p, z)$. Here y is replaced by y/p, and the sum over b of $\mathbf{e}(b(r^2-\lambda^2)/4p)$ equals p if p divides the integer $(r^2-\lambda^2)/4$ and 0 otherwise. Write $r = \lambda + 2t$, so that $(r^2-\lambda^2)/4 = t(\lambda+t)$. For every prime p (including $p = 2$), the condition $p \mid t(\lambda+t)$ together with $r \equiv \lambda \pmod 2$ means $r \equiv \lambda$ or $r \equiv -\lambda \pmod{2p}$, and both hold exactly when $p \mid \lambda$ and $r \equiv \lambda \pmod{2p}$. Since the summand depends on λ only through λ^2, the sum over the union of the two congruence conditions is twice the sum over $r \equiv \lambda \pmod{2p}$ minus the sum over $p \mid \lambda$, $r \equiv \lambda \pmod{2p}$. So these terms equal

$$\frac{\sqrt{p}}{8\pi\sqrt{y}} \Big(2 \sum_{\lambda} \sum_{r \equiv \lambda\,(2p)} - \sum_{p|\lambda} \sum_{r \equiv \lambda\,(2p)}\Big) I\Big(\frac{\pi\lambda^2 y}{p}\Big) \mathbf{e}\Big(\frac{(r^2-\lambda^2)\tau}{4p}\Big) \zeta^r.$$

The subtracted part cancels the term with $a = p$, and what remains is

$$\frac{2\sqrt{p}}{8\pi\sqrt{y}} \sum_{l \bmod 2p} \sum_{\lambda \equiv l\,(2p)} I\Big(\frac{\pi\lambda^2 y}{p}\Big) q^{-\lambda^2/4p}\, \vartheta_{p,l} = \frac{\sqrt{p}}{4\pi} \sum_{l \bmod 2p} S_{p,l}\, \vartheta_{p,l}.$$

□

The factor 2 in Lemma 5.4 has no counterpart for $m = 1$: comparing with Lemma 5.2, the nonholomorphic part of $\widehat{\mathcal{H}}$ is $\frac{1}{8\pi} \sum S_{1,\mu} \vartheta_{1,\mu}$, half that of $\widehat{A}_{1,1}$, while for a prime p the nonholomorphic parts of $\widehat{\mathcal{H}}|V_p$ and $\widehat{A}_{1,p}$ coincide. This is why $\lambda_1 = 2p_{24}(2)$ and $\lambda_p = p_{24}(p+1)$.

5.3. **Proof of Theorem D.** Let $m = 1$ or let m be prime. By (4),

$$\phi_m = \Delta\psi_m^F + \lambda_m \mathcal{H}|V_m = \frac{\chi_{m+1}}{A} - p_{24}(m+1) A_{1,m} + \lambda_m \mathcal{H}|V_m.$$

By Lemmas 5.2, 5.3 and 5.4,

$$-p_{24}(m+1)A_{1,m} + \lambda_m \mathcal{H}|V_m = -p_{24}(m+1)\widehat{A}_{1,m} + \lambda_m \widehat{\mathcal{H}}|V_m + \kappa_m \sum_{l \bmod 2m} S_{m,l} \vartheta_{m,l},$$

with $\kappa_1 = \frac{p_{24}(2)}{4\pi} - \frac{\lambda_1}{8\pi} = 0$ and $\kappa_p = \frac{\sqrt{p}}{4\pi}(p_{24}(p+1) - \lambda_p) = 0$. Therefore $\phi_m = \chi_{m+1}/A - p_{24}(m+1)\widehat{A}_{1,m} + \lambda_m \widehat{\mathcal{H}}|V_m$, and each of the three terms satisfies (E) and (M) with weight 2 and index m: the first because χ_{m+1} and A are weak Jacobi forms of weights $0, -2$ and indices $m+1, 1$, the second by Lemma 5.2, and the third by Lemma 5.3 and [14, Thm. 4.2]. On the other hand ϕ_m is holomorphic in τ and in z, because ψ_m^F is (Lemma 4.2) and $\mathcal{H}|V_m$ is a theta series with

polynomially bounded coefficients ($H(N) \le N+1$ by Lemma C.1). Its expansion has $n \ge 0$, since ψ_m^F has $n \ge -1$ and $\mathcal{H}|V_m$ has $n \ge 0$. So ϕ_m is a weak Jacobi form of weight 2 and index m.

By (8), $\phi_m = \sum_i x_i M_i$ for the monomials $M_i = E_4^\alpha E_6^\beta A^i (2\phi_{0,1})^{m-i}$ with $4\alpha + 6\beta - 2i = 2$, $1 \le i \le m$; there are $1, 2, 3, 6, 9, 18, 23$ of them for $m = 1, 2, 3, 5, 7, 11, 13$. The identity $\Delta\psi_m^F = \sum_i x_i M_i - \lambda\,\mathcal{H}|V_m$ is linear in the unknowns (x_i, λ). We computed the coefficients of $q^n\zeta^l$, $0 \le n \le 14$, $0 \le l \le m$, of $\Delta\psi_m^F$, of the M_i and of $\mathcal{H}|V_m$ in exact rational arithmetic (Appendix B), and found that the resulting matrix has full column rank and that the system has the solution listed in Appendix A, with $\lambda = \lambda_m$. Since a solution exists by the first part of the proof and the rank is full, the solution is unique, and the x_i are the coefficients of ϕ_m. □

Remark 5.5. For composite m the argument fails at Lemma 5.4, and so does the identity: for $m = 4$ and $m = 6$ the same linear system, with the coefficients $0 \le n \le 12$, has no solution. Some other combination of class number generating functions is then needed, and Lemma 4.3 fails as well. This is why Theorem B is restricted to $m = 1$ and prime m.

6. The exact formula

Let $m \in \mathcal{M}$. By Theorem D, ϕ_m/Δ is a weakly holomorphic Jacobi form of weight -10 and index m. We write its coefficients as $a_m(n,l)$, and, since they depend only on $D = 4mn - l^2$ and on l mod $2m$, also as $a_m(D,l)$. Let $\delta_m(n,l)$ be the coefficient of $q^n\zeta^l$ in $\mathcal{H}|V_m/\Delta$, so that

$$\delta_m(n,l) = \sum_{j\ge 0} p_{24}(j)\, h_m(n+1-j,l), \qquad c_m^F(n,l) = a_m(n,l) - \lambda_m\,\delta_m(n,l), \tag{18}$$

with h_m as in (7). The pairs (D',l') with $D' < 0$ and $a_m(D',l') \ne 0$ are the *polar terms*; there are finitely many classes of them, because $a_m(n,l) = 0$ for $n < -1$ and the coefficients of a weak Jacobi form vanish for $4mn - l^2 < -m^2$. The most polar one is $D' = -m(m+4)$, $l' \equiv m$, where $a_m = c_m^F(-1,m) = m+2$ by Lemma 4.4, since $h_m(0,m) = H(-m^2) = 0$. The other polar coefficients of ϕ_m/Δ differ from those of ψ_m^F only at $(D',l') = (-4m, 0)$, where $\delta_m(-1,0) = \sigma(m)H(0) = -\sigma(m)/12$; all of them are listed in Appendix A.

By (9) and (10), the functions $g_l(\tau) = \sum_D a_m(D,l) q^{D/4m}$, $l \in \mathbb{Z}/2m\mathbb{Z}$, form a vector-valued weakly holomorphic modular form of weight $w = -\frac{21}{2}$ for $\mathrm{SL}_2(\mathbb{Z})$ (more precisely for its metaplectic double cover), with a representation ρ determined by (10). The matrices in (10) are unitary: the first is diagonal with entries of modulus one, and the second is a scalar of modulus one times $(2m)^{-1/2}(\mathbf{e}(lj/2m))_{l,j}$, whose rows are orthonormal. Hence ρ is unitary. Put $\nu = 1 - w = \frac{23}{2}$. Exact formulae of the following kind for the coefficients of modular forms of negative weight go back to Rademacher and Zuckerman [19].

Theorem 6.1. *For $D > 0$ with $D \equiv -l^2 \pmod{4m}$,*

$$a_m(D,l) = \sum_{c\ge 1} \frac{2\pi}{c} \sum_{(D',l')} a_m(D',l')\, K_c(D',l';D,l) \Big(\frac{|D'|}{D}\Big)^{\nu/2} I_\nu\Big(\frac{\pi\sqrt{|D'|D}}{mc}\Big), \tag{19}$$

where (D',l') runs over the polar terms with $l' \in \mathbb{Z}/2m\mathbb{Z}$, I_ν is the modified Bessel function of the first kind, and the Kloosterman sum $K_c(D',l';D,l)$ is a sum of $\varphi(c)$ complex numbers of modulus at most one. For $c = 1$, $K_1(D',l';D,l) = -(2m)^{-1/2}\mathbf{e}(ll'/2m)$. The series converges absolutely.

Proof. For vector-valued modular forms of negative weight with a unitary representation, Bruinier constructs for each polar term a Maass–Poincaré series of weight w whose principal part is $q^{D'/4m}(\mathfrak{e}_{l'} + \mathfrak{e}_{-l'})$, where $\mathfrak{e}_l$ is the standard basis vector, and computes its Fourier expansion

[7, Def. 1.8, Prop. 1.10]. Its holomorphic part has, for $D > 0$, the coefficients given by the sum over c in (19), where $K_c(D', l'; D, l)$ is a sum over d mod c with $(c, d) = 1$ of a root of unity times an entry of $\rho(\gamma)^{-1}$, for a matrix $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with lower row (c, d) [7, Prop. 1.10]; Manschot and Moore give the same series in the notation of Jacobi forms [18, §§2–3]. Since ρ is unitary, each term of K_c has modulus at most one. For $c = 1$ there is one term, with $\gamma = S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and the entry of $\rho(S)$ given by (10) with $k = -10$ yields $K_1 = -(2m)^{-1/2}\mathbf{e}(\pm ll'/2m)$; the sign of the exponent is immaterial because $a_m(D', l') = a_m(D', -l')$ and the sum in (19) contains both l' and $-l'$.

Let P be the sum over the polar terms of $a_m(D', l')$ times the corresponding Poincaré series (each pair $\pm l'$ taken once). Then $h = (g_l)_l - P$ transforms with weight w and representation ρ, is annihilated by the weight w Laplacian, and has no principal part. Its image $\xi_w(h)$ under the antilinear operator $\xi_w = 2iy^w\overline{\partial_{\bar\tau}}$ is a cusp form of weight $2 - w$ for the dual representation [8, Prop. 3.2]. By [8, Prop. 3.5], the Petersson norm of $\xi_w(h)$ equals a finite sum of products $c_h^+(n)\, b(-n)$ over $n \le 0$, where $c_h^+(n)$ are the coefficients of the holomorphic part of h and $b(n)$ those of $\xi_w(h)$. For $n < 0$ the first factor vanishes because h has no principal part, and for $n = 0$ the second vanishes because $\xi_w(h)$ is a cusp form; so $\xi_w(h) = 0$. Hence h is a weakly holomorphic modular form of negative weight without principal part, that is, a holomorphic modular form of negative weight, and therefore $h = 0$. So $(g_l) = P$, and comparing holomorphic parts gives (19).

Absolute convergence: $I_\nu(t) \le (t/2)^\nu e^t/\Gamma(\nu + 1)$ (Lemma 7.1), so the term of index c is $O(c^{-1}\varphi(c)c^{-\nu}) = O(c^{-\nu})$ and $\nu > 1$. □

As a check on the normalisation, the terms with $c = 1$ of (19) reproduce the exact coefficients $a_m(n, l)$ computed from the definitions to a relative accuracy better than $2 \cdot 10^{-25}$ at the largest n in our tables, for every $m \in \mathcal{M}$ and $0 \le l \le m$ (Appendix B).

6.1. **Main term and error terms.** For (D', l') a polar term with $0 \le l' \le m$, put $w_{l'} = 1$ if $l' \in \{0, m\}$ and $w_{l'} = 2$ otherwise; this is the number of classes $\pm l'$ modulo $2m$. Let

$$x_{D'}(D) = \frac{\pi\sqrt{|D'|D}}{m}, \qquad \alpha = \alpha_m = \pi\sqrt{\frac{m+4}{m}},$$

so that the most polar term has $x_{-m(m+4)}(D) = \alpha\sqrt{D}$. The main term is

$$T(D) = T_m(D) = \frac{2\pi}{\sqrt{2m}}\,(m+2)\Big(\frac{m(m+4)}{D}\Big)^{\nu/2} I_\nu(\alpha\sqrt{D}). \tag{20}$$

The terms with $c = 1$ of (19) are

$$-\frac{2\pi}{\sqrt{2m}} \sum_{(D', l')} w_{l'}\, a_m(D', l') \cos\Big(\frac{\pi ll'}{m}\Big)\Big(\frac{|D'|}{D}\Big)^{\nu/2} I_\nu\big(x_{D'}(D)\big),$$

now with $0 \le l' \le m$. For the most polar term, $\cos(\pi lm/m) = (-1)^l$, so its contribution to $F_m(D) = (-1)^{l+1}c_m^F(n, l)$ is exactly $T(D)$.

Proposition 6.2. *Let $m \in \mathcal{M}$, let $D \ge 4m$ with $D \equiv -l^2 \pmod{4m}$, and let $\mathcal{S}$ be any set of polar terms. Then*

$$|F_m(D) - T(D)| \le E_1(D) + E_2(D) + E_3(D),$$

where, with $B_1(t) = e^t/\sqrt{2\pi t}$, $B_2(t) = (t/2)^\nu e^t/\Gamma(\nu+1)$ *and* $Z = \sum_{c\ge 2} c^{-\nu}$,

$$E_1(D) = \frac{2\pi}{\sqrt{2m}} \sum_{(D',l')\neq(-m(m+4),m)} w_{l'}|a_m(D',l')|\Big(\frac{|D'|}{D}\Big)^{\nu/2} B_{(D',l')}\big(x_{D'}(D)\big),$$

$$E_2(D) = \frac{2\pi Z}{\Gamma(\nu+1)} \sum_{(D',l')} w_{l'}|a_m(D',l')|\Big(\frac{|D'|}{D}\Big)^{\nu/2}\Big(\frac{x_{D'}(D)}{2}\Big)^{\nu} e^{x_{D'}(D)/2},$$

$$E_3(D) = \lambda_m\,\sigma(m)\,(D+4m+1)\Big(25 + c_0\Big(\frac{D}{4m}\Big)^{-5} e^{2\pi\sqrt{D/m}}\Big),$$

$B_{(D',l')} = B_2$ *for* $(D',l') \in \mathcal{S}$ *and* B_1 *otherwise,* $\sigma(m)$ *is the sum of the divisors of* m, *and* $c_0 = \prod_{k\ge1}(1-e^{-2\pi k})^{-24} < 1.046$.

Proof. By (18), $F_m(D) = (-1)^{l+1}a_m(D,l) - (-1)^{l+1}\lambda_m\delta_m(n,l)$. In (19), the terms with $c=1$ other than the main term are bounded in absolute value by $E_1(D)$, because $|\cos| \le 1$ and $I_\nu \le B_1$, $I_\nu \le B_2$ (Lemma 7.1). For $c \ge 2$ we have $c^{-1}|K_c| \le c^{-1}\varphi(c) \le 1$, and by the series of I_ν, $I_\nu(t/c) \le (t/2c)^\nu e^{t/c}/\Gamma(\nu+1) \le c^{-\nu}(t/2)^\nu e^{t/2}/\Gamma(\nu+1)$; summing over $c \ge 2$ and over the classes gives $E_2(D)$. The bound $\lambda_m|\delta_m(n,l)| \le E_3(D)$ is Lemma 6.3. □

Lemma 6.3. *For* $D = 4mn - l^2 \ge 4m$ *and* $m \in \mathcal{M}$, $|\delta_m(n,l)| \le \sigma(m)(D+4m+1)\big(25 + c_0(D/4m)^{-5}e^{2\pi\sqrt{D/m}}\big)$.

Proof. The term j of (18) involves $h_m(k,l)$ with $k = n+1-j$, whose discriminant is $4mk-l^2 = D+4m(1-j)$. It vanishes unless this is nonnegative, that is unless $j \le J$ with $J = 1+D/4m$. For such j, (7) and Lemma C.1 give $|h_m(k,l)| \le \sum_{d|m} d\,(D+4m+1) = \sigma(m)(D+4m+1)$. So

$$|\delta_m(n,l)| \le \sigma(m)(D+4m+1)\sum_{0\le j\le J} p_{24}(j).$$

Here $p_{24}(0)+p_{24}(1) = 25$, and for $2 \le j \le J$ Lemma C.2 gives $p_{24}(j) \le c_0\,t^{-6}e^{4\pi\sqrt{t}}$ with $t = j-1 \in [1, J-1]$. The function $t^{-6}e^{4\pi\sqrt{t}}$ is increasing for $t \ge 1$, since its logarithmic derivative $-6/t + 2\pi/\sqrt{t}$ is positive for $t > 9/\pi^2$. There are at most $J-1$ such j, so their sum is at most $(J-1)c_0(J-1)^{-6}e^{4\pi\sqrt{J-1}}$, and $J-1 = D/4m \ge 1$, $4\pi\sqrt{D/4m} = 2\pi\sqrt{D/m}$. □

7. Bessel functions

We use the integral representation [25, §3.71]

$$I_\nu(x) = \frac{(x/2)^\nu}{\sqrt{\pi}\,\Gamma(\nu+\frac12)}\int_{-1}^{1}(1-t^2)^{\nu-\frac12}e^{xt}\,dt \qquad (\nu > -\tfrac12,\ x>0), \tag{21}$$

the series $I_\nu(x) = \sum_{k\ge0}(x/2)^{\nu+2k}/(k!\,\Gamma(\nu+k+1))$, and the recurrences $I'_\nu = I_{\nu+1} + \frac{\nu}{x}I_\nu$, $I'_{\nu+1} = I_\nu - \frac{\nu+1}{x}I_{\nu+1}$ [25, §3.71]. Substituting $t = 1-u$ in (21) gives

$$I_\nu(x) = \frac{(x/2)^\nu e^x}{\sqrt{\pi}\,\Gamma(\nu+\frac12)}\int_0^2 u^{\nu-\frac12}(2-u)^{\nu-\frac12}e^{-xu}\,du. \tag{22}$$

Lemma 7.1. *For* $\nu \ge \frac12$ *and* $x > 0$, $I_\nu(x) \le e^x/\sqrt{2\pi x}$. *For* $\nu \ge 0$ *and* $x > 0$, $I_\nu(x) \le (x/2)^\nu e^x/\Gamma(\nu+1)$.

Proof. For $\nu \ge \frac{1}{2}$, bound $(2-u)^{\nu-\frac{1}{2}} \le 2^{\nu-\frac{1}{2}}$ in (22) and extend the integral to $(0,\infty)$:

$$I_\nu(x) \le \frac{(x/2)^\nu e^x}{\sqrt{\pi}\,\Gamma(\nu+\frac{1}{2})}\, 2^{\nu-\frac{1}{2}}\,\Gamma(\nu+\tfrac{1}{2})\, x^{-\nu-\frac{1}{2}} = \frac{e^x}{\sqrt{2\pi x}}.$$

For the second bound, $\Gamma(\nu+k+1) = \Gamma(\nu+1)(\nu+1)\cdots(\nu+k) \ge \Gamma(\nu+1)\,k!$, so

$$I_\nu(x) \le \frac{(x/2)^\nu}{\Gamma(\nu+1)} \sum_{k\ge 0} \frac{(x/2)^{2k}}{k!^2} = \frac{(x/2)^\nu}{\Gamma(\nu+1)}\, I_0(x),$$

and $I_0(x) = \frac{1}{\pi}\int_{-1}^{1}(1-t^2)^{-1/2}e^{xt}dt \le e^x$ by (21) with $\nu = 0$. □

Lemma 7.2. *Let $\nu = \frac{23}{2}$ and $0 < s < 2$. For $x > 0$,*

$$I_\nu(x) \ge \frac{e^x}{\sqrt{2\pi x}}\Big(1-\frac{s}{2}\Big)^{11} P(12, sx), \qquad P(12,t) = \frac{1}{11!}\int_0^t v^{11}e^{-v}\,dv.$$

Proof. In (22), $\nu - \frac{1}{2} = 11$. Keep only $0 \le u \le s$, where $2-u \ge 2(1-s/2)$:

$$\int_0^2 u^{11}(2-u)^{11}e^{-xu}du \ge 2^{11}\Big(1-\frac{s}{2}\Big)^{11}\int_0^s u^{11}e^{-xu}du = 2^{11}\Big(1-\frac{s}{2}\Big)^{11} x^{-12}\,11!\,P(12,sx).$$

Since $\Gamma(\nu+\frac{1}{2}) = 11!$, this gives $I_\nu(x) \ge (x/2)^{23/2}e^x\pi^{-1/2}2^{11}x^{-12}(1-s/2)^{11}P(12,sx)$, and $(x/2)^{23/2}2^{11}x^{-12}\pi^{-1/2} = (2\pi x)^{-1/2}$. □

We use $s = \frac{1}{5}$. The factor $\kappa(x) = (1-\frac{s}{2})^{11}P(12,sx)$ is increasing in x and tends to $0.9^{11} = 0.3138\ldots$ as $x \to \infty$.

Lemma 7.3. *Let $\nu \ge 0$, $a = \nu+1$ and $g(x) = x/(a+\sqrt{x^2+a^2})$. Then $I_{\nu+1}(x)/I_\nu(x) > g(x)$ for all $x > 0$.*

Proof. This is Amos' inequality [1]; we include a proof. Let $R = I_{\nu+1}/I_\nu$. The recurrences give

$$R' = \frac{I'_{\nu+1}}{I_\nu} - R\frac{I'_\nu}{I_\nu} = 1 - R^2 - \frac{2\nu+1}{x}R.$$

Put $\Phi(x,r) = 1 - r^2 - (2\nu+1)r/x$ and $\sigma = \sqrt{x^2+a^2}$, so that $g = (\sigma-a)/x$. A direct computation gives

$$\Phi(x,g) - g'(x) = \frac{(\sigma-a)^2}{x^2\sigma} = \frac{\sigma-a}{\sigma(\sigma+a)} > 0.$$

As $x \to 0^+$ the series give $R = \frac{x}{2a}\big(1-\frac{x^2}{4a(a+1)}\big)+O(x^5)$ and $g = \frac{x}{2a}\big(1-\frac{x^2}{4a^2}\big)+O(x^5)$, so $R > g$ on some interval $(0,x_0)$. If $R-g$ had a first zero $x_1 > 0$, then $(R-g)'(x_1) \le 0$; but at x_1,

$$(R-g)'(x_1) = \Phi(x_1,R) - g'(x_1) = \Phi(x_1,g) - g'(x_1) > 0.$$

Hence $R > g$ for all $x > 0$. □

Lemma 7.4. *Let $m \in \mathcal{M}$, $a = \nu+1 = \frac{25}{2}$, and put $\beta(D) = g(\alpha\sqrt{D})$ and*

$$\gamma(D) = \gamma_m(D) = 1 - \exp\Big(-\frac{2\alpha\beta(D)}{\sqrt{D+4}}\Big).$$

Then T is increasing and $T(D) \le (1-\gamma(D))\,T(D+4)$ for all $D > 0$. Moreover $\sqrt{D}\,\gamma(D)$ is increasing for $D \ge 36$.

Proof. From (20) and $I'_\nu = I_{\nu+1} + \frac{\nu}{x} I_\nu$, with $x = \alpha\sqrt{D}$,

$$\frac{d}{dD}\log T(D) = -\frac{\nu}{2D} + \frac{\alpha}{2\sqrt{D}}\frac{I'_\nu(x)}{I_\nu(x)} = \frac{\alpha}{2\sqrt{D}}\,\frac{I_{\nu+1}(x)}{I_\nu(x)} > \frac{\alpha\, g(\alpha\sqrt{D})}{2\sqrt{D}} > 0,$$

because $\nu/(2D) = \frac{\alpha}{2\sqrt{D}} \cdot \frac{\nu}{x}$. The function g is increasing, so for $D \le t \le D+4$ the integrand is at least $\alpha\beta(D)/(2\sqrt{D+4})$, and integrating over $[D, D+4]$ gives $\log T(D+4) - \log T(D) \ge 2\alpha\beta(D)/\sqrt{D+4}$.

For the last statement write $\gamma = 1 - e^{-u}$ with $u = 2\alpha\beta/\sqrt{D+4}$, so that $\sqrt{D}\gamma = (\sqrt{D}u)\cdot \frac{1-e^{-u}}{u}$. The first factor $2\alpha\beta\sqrt{D/(D+4)}$ is increasing. The second is a decreasing function of u, so it suffices to show that u decreases. From $g = (\sigma - a)/x$ with $\sigma = \sqrt{x^2 + a^2}$ one finds $\frac{d}{dx}\log g = \frac{x}{\sigma(\sigma-a)} - \frac{1}{x} = \frac{a}{x\sigma}$, hence

$$\frac{d}{dD}\log u = \frac{a}{x\sigma}\cdot\frac{x}{2D} - \frac{1}{2(D+4)} = \frac{1}{2D(D+4)\sigma}\big(a(D+4) - D\sigma\big),$$

which is negative when $\sigma > a(1 + 4/D)$. Since $\alpha \ge \pi$, for $D \ge 36$ we have $\sigma > \alpha\sqrt{D} \ge 6\pi > 13.9 > a(1+4/D)$. □

8. The tail estimate

Let $m \in \mathcal{M}$ and let D^*_m be the number in Table 1. In Proposition 6.2 take for $\mathcal{S}$ the polar terms with $x_{D'}(D^*_m) < 12$, so that the bound B_2 is used for the terms whose Bessel argument is small and B_1 for the others; any choice of $\mathcal{S}$ gives a valid bound, and this one is fixed once and for all. Let $T_{\mathrm{low}}(D)$ be (20) with $I_\nu(\alpha\sqrt{D})$ replaced by the lower bound of Lemma 7.2 with $s = \frac{1}{5}$, and put

$$\varepsilon(D) = \varepsilon_m(D) = \frac{E_1(D) + E_2(D) + E_3(D)}{T_{\mathrm{low}}(D)}.$$

By Proposition 6.2 and Lemma 7.2, for $D \ge 4m$,

$$|F_m(D) - T(D)| \le \varepsilon(D)\, T_{\mathrm{low}}(D) \le \varepsilon(D)\, T(D). \tag{23}$$

Lemma 8.1. *Let $a_0 = \alpha$, $a_{D'} = \pi\sqrt{|D'|}/m$, and let Θ_m be the largest of the numbers*

$$\frac{2e_{D'}}{a_0 - a_{D'}}\ \ (D' \ne -m(m+4)), \qquad \frac{13}{a_0 - a_{D'}/2}, \qquad \frac{15}{a_0}, \qquad \frac{5}{a_0 - 2\pi/\sqrt{m}},$$

over the polar terms, where $e_{D'} = \frac{13}{2}$ for terms in $\mathcal{S}$ and $e_{D'} = \frac{1}{2}$ otherwise. If all denominators are positive, then $\sqrt{D}\,\varepsilon(D)$ is nonincreasing for $\sqrt{D} > \Theta_m$.

Proof. Since $\kappa(\alpha\sqrt{D})$ increases with D, it suffices to show that $\sqrt{D}$ times each summand of $E_1 + E_2 + E_3$, divided by $D^{-\nu/2}(\alpha\sqrt{D})^{-1/2}e^{\alpha\sqrt{D}}$, is nonincreasing. Up to positive constants these quotients are:

- a term of E_1 with B_1: $D^{1/2}e^{-(a_0 - a_{D'})\sqrt{D}}$, since the factors $D^{-\nu/2}$ and $D^{-1/4}$ cancel;
- a term of E_1 with B_2: $D^{\nu/2+3/4}e^{-(a_0-a_{D'})\sqrt{D}} = D^{13/2}e^{-(a_0-a_{D'})\sqrt{D}}$, since $x_{D'}(D)^\nu$ cancels $D^{-\nu/2}$;
- a term of E_2: $D^{13/2}e^{-(a_0-a_{D'}/2)\sqrt{D}}$;
- the first part of E_3: $(D+4m+1)D^{\nu/2+3/4}e^{-a_0\sqrt{D}}$;
- the second part of E_3: $(D+4m+1)D^{-5}D^{\nu/2+3/4}e^{-(a_0-2\pi/\sqrt{m})\sqrt{D}} = (D+4m+1)D^{3/2}e^{-(a_0-2\pi/\sqrt{m})\sqrt{D}}$.

The logarithmic derivative of $D^e e^{-b\sqrt{D}}$ is $e/D - b/(2\sqrt{D})$, which is negative for $\sqrt{D} > 2e/b$, and $\frac{d}{dD}\log(D+4m+1) \le 1/D$, which raises the exponent e by one in the last two cases: $e = \frac{15}{2}$ and $e = \frac{5}{2}$. These are the five conditions defining Θ_m. □

Proposition 8.2. *For each $m \in \mathcal{M}$, the numbers $\varepsilon_m(D_m^*)$ and $\gamma_m(D_m^*)$ lie in the intervals of Table 1, all denominators in Lemma 8.1 are positive, $\sqrt{D_m^*} > \Theta_m$, $\varepsilon_m(D_m^*) < \frac{1}{2}$, and $2\varepsilon_m(D_m^*) < \gamma_m(D_m^*)$.*

Proof. The quantities T_{low}, E_1, E_2, E_3, γ and Θ_m are explicit finite expressions in π, square roots, exponentials, $\Gamma(\nu+1)$, the regularised incomplete gamma function $P(12,\cdot)$, the polar coefficients of Appendix A, and the constants Z and c_0. We evaluated them in ball arithmetic with the Arb library [17] at 200 bits, which returns intervals guaranteed to contain the exact values. For Z we used $\sum_{c=2}^{399} c^{-\nu} + \int_{399}^{\infty} t^{-\nu}dt$, and for c_0 the product over $k < 60$ multiplied by $1 + 10^{-100}$, which exceeds the remaining factors since $\prod_{k\ge 60}(1-e^{-2\pi k})^{-24} < \exp(25e^{-120\pi})$. The results are in Table 1. □

TABLE 1. The tail certificate at $D = D_m^*$. The columns give the three error terms relative to T_{low}, their sum ε, the gap γ of Lemma 7.4, and the threshold Θ_m of Lemma 8.1, to be compared with $\sqrt{D_m^*}$. All entries are rounded outward from Arb intervals.

m	D_m^*	E_1/T_{low}	E_2/T_{low}	E_3/T_{low}	ε_m	γ_m	Θ_m	$\sqrt{D_m^*}$
1	680	$7.46\cdot10^{-9}$	$2.38\cdot10^{-27}$	0.1962	0.19612	0.39455	6.743	26.07
2	371	$1.92\cdot10^{-5}$	$4.37\cdot10^{-13}$	0.1932	0.19289	0.39282	5.008	19.26
3	292	$9.17\cdot10^{-4}$	$2.82\cdot10^{-9}$	0.1855	0.18543	0.38075	5.419	17.08
5	258	0.04356	$6.42\cdot10^{-7}$	0.1292	0.17271	0.35171	6.170	16.06
7	313	0.1445	$4.38\cdot10^{-7}$	0.00919	0.15369	0.30931	6.603	17.69
11	720	0.1074	$1.68\cdot10^{-11}$	$4.22\cdot10^{-9}$	0.10667	0.21356	7.088	26.83
13	1030	0.09065	$2.67\cdot10^{-14}$	$4.37\cdot10^{-13}$	0.090641	0.18169	7.238	32.09

Theorem 8.3 (Tail lemma)**.** *Let $m \in \mathcal{M}$, and let D, D' be discriminants of charges of index m with $D_m^* \le D' \le D-4$. Then $F_m(D) > F_m(D') > 0$.*

Proof. Write $\varepsilon' = \varepsilon(D')$ and $\gamma' = \gamma(D')$. By Lemma 8.1 and Proposition 8.2, $\sqrt{t}\,\varepsilon(t)$ is nonincreasing for $t \ge D_m^*$; hence ε is nonincreasing there, $\varepsilon(D) \le \varepsilon' \le \varepsilon(D_m^*) < \frac{1}{2}$, and, using Lemma 7.4,

$$2\varepsilon' \le 2\varepsilon(D_m^*)\sqrt{D_m^*/D'} < \gamma(D_m^*)\sqrt{D_m^*/D'} \le \gamma'.$$

By (23), $F_m(D') \ge (1-\varepsilon')T(D') > 0$. By Lemma 7.4 and the monotonicity of T, $T(D') \le (1-\gamma')T(D'+4) \le (1-\gamma')T(D)$. Therefore

$$F_m(D) - F_m(D') \ge (1-\varepsilon(D))T(D) - (1+\varepsilon')(1-\gamma')T(D) \ge \big(\gamma'(1+\varepsilon') - 2\varepsilon'\big)T(D) > 0. \ \square$$

Figure 2 shows ε_m against $\gamma_m/2$ on both sides of D_m^*, and Figure 3 compares $F_m(D)$ with the main term $T_m(D)$.

9. Proof of Theorem B and Corollary C

Let $m \in \mathcal{M}$. Define

$$M_m = \max\Big(\max_{\substack{c\le a\le m,\ 0\le b\le c\\ 4ac-b^2>0}} (-1)^{b+1} c_c^F(a,b),\quad \max_{0<D'<D_m^*} F_m(D')\Big),$$

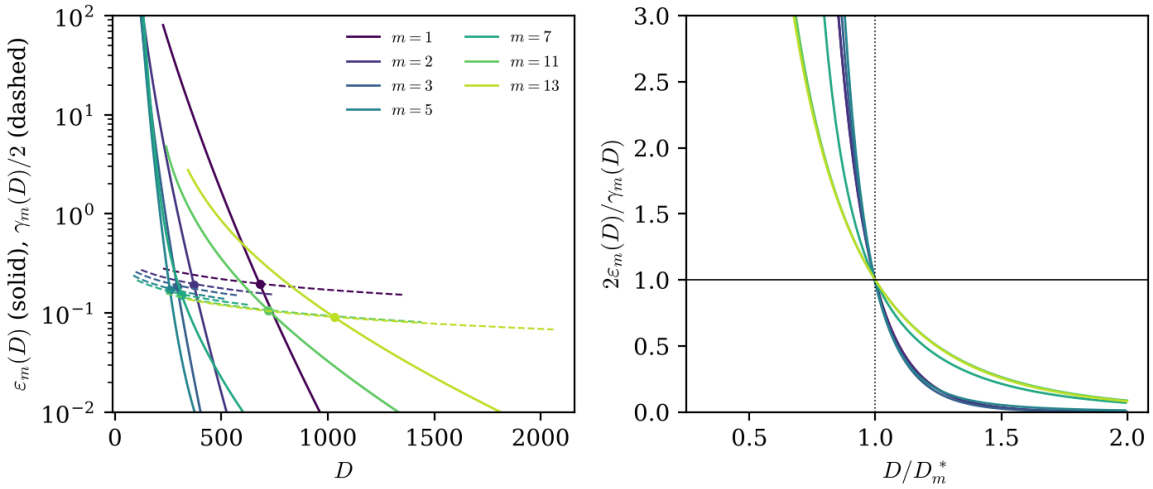


FIGURE 2. Left: the relative error bound $\varepsilon_m(D)$ (solid) and half the relative gap $\gamma_m(D)/2$ (dashed) for $m \in \mathcal{M}$; the dots mark $\varepsilon_m(D_m^*)$. Right: the ratio $2\varepsilon_m/\gamma_m$ against D/D_m^*. The tail lemma needs this ratio below 1 at D_m^*; Lemmas 7.4 and 8.1 then keep it below 1 for all larger D. For $m = 1, 2, 3$ the class number term E_3 dominates the error, for $m = 7, 11, 13$ the second polar term does.

the second maximum over discriminants of charges of index m, and let $D_1 \geq D_m^*$ be the smallest such discriminant with $F_m(D_1) > M_m$. These numbers are in Table 2.

TABLE 2. Data for the proof of Theorem B. The last two columns count the charges with $1000 < D < D_1+4$ checked directly, and the reduced forms $[a, b, m]$ with $0 < D < D_m^*$ checked for Corollary C.

m	D_m^*	M_m	D_1	$F_m(D_1)/M_m$	direct charges	reduced forms
1	680	$4.687 \cdot 10^{66}$	680	1.134	0	339
2	371	$1.586 \cdot 10^{36}$	372	1.656	0	135
3	292	$2.446 \cdot 10^{28}$	296	1.831	0	89
5	258	$2.241 \cdot 10^{24}$	259	1.396	0	53
7	313	$1.295 \cdot 10^{26}$	315	1.440	0	43
11	720	$3.019 \cdot 10^{38}$	723	1.682	0	82
13	1030	$5.873 \cdot 10^{45}$	1031	1.199	148	118

Proof of Theorem B. Let n and $J \geq 0$ be integers with $D = 4mn - J^2 > 0$. If $D \leq 1000$, then $d(n, J, m) > 0$ by Theorem A. Let $D > 1000$. Since $4m^2 \leq 676 < D$, Lemma 3.3 shows that the reduced form of (n, J, m) has $c = m$, so $d^*(n, J, m) = F_m(D)$ by Lemma 4.3. Put $D' = 4mn - (J+2)^2 = D - 4(J+1)$, the discriminant of the second charge.

Case $D' \leq 0$. Then $d^*(n, J+2, m) = 0$ and $d(n, J, m) = d^*(n, J, m)$, which is positive by Corollary C; its proof below does not use Theorem B.

Case $D' \geq D_m^*$. Then $D' > 4m^2$, so the reduced form of $(n, J+2, m)$ also has $c = m$ by Lemma 3.3, and $d^*(n, J+2, m) = F_m(D')$. Since $D' \leq D - 4$, Theorem 8.3 gives $F_m(D) > F_m(D')$.

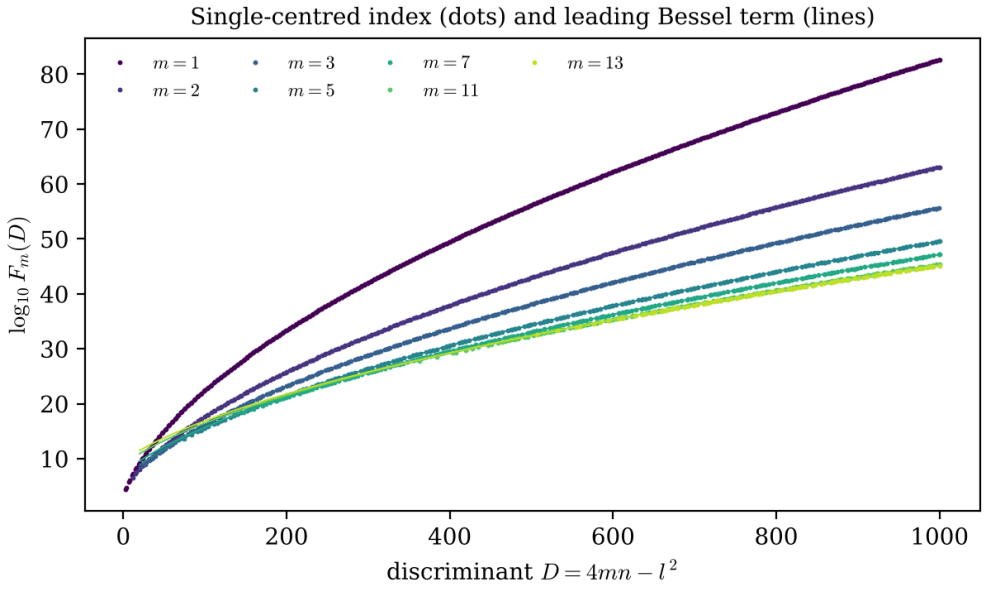


FIGURE 3. The single-centred index $F_m(D)$ at the discriminants of reduced charges with $D \le 1000$ (dots), and the main term $T_m(D)$ of (20) (lines), on a logarithmic scale. At $D = 1000$ the index has between 46 digits ($m = 11, 13$) and 83 digits ($m = 1$); the relative difference between dots and lines is below $\varepsilon_m(D)$ once $D \ge D_m^*$.

Case $0 < D' < D_m^*$. By Lemma 3.3, the reduced form (a, b, c) of $(n, J+2, m)$ has $c = m$, and then $d^*(n, J+2, m) = F_m(D') \le M_m$, or it has $c \le a \le m$, and then $d^*(n, J+2, m) = (-1)^{b+1} c_c^F(a, b) \le M_m$. If $D \ge D_1 + 4$, Theorem 8.3 gives $F_m(D) > F_m(D_1) > M_m \ge d^*(n, J+2, m)$. If $1000 < D < D_1 + 4$, which happens only for $m = 13$, then $J < D/4$ and the charge is among those checked directly.

In every case $d(n, J, m) > 0$. □

The direct check for $m = 13$ computes $d(n, J, 13)$ exactly for the 148 charges with $1000 < D < 1035$ and $0 \le J \le D/4$ from the tables of Appendix B, using reduced forms for both terms; all values are positive.

Proof of Corollary C. Let (a, b, c) be the reduced form of (n, l, m'), with $c \in \mathcal{M}$ and $D > 0$. By (5) and Lemma 4.3, $d^*(n, l, m') = (-1)^{b+1} c_c^F(a, b) = F_c(D)$. If $D \ge D_c^*$, this is positive by (23), because $\varepsilon_c(D) < \frac{1}{2}$ by Proposition 8.2 and Lemma 8.1. If $D < D_c^*$, the reduced form $[a, b, c]$ is one of the forms counted in the last column of Table 2, and all of them have $(-1)^{b+1} c_c^F(a, b) > 0$. □

Figure 4 shows the index $d(n, J, 7)$ over the region $D \le 1000$.

10. Proof of Theorem A

Four dimensions. Every charge with $0 < D \le 1000$ has a reduced form (a, b, c) with $0 < 4ac - b^2 \le 1000$, and then $3c^2 \le D \le 1000$ by Lemma 3.1(iv), so $c \le 18$. We computed $c_c^F(a, b)$ exactly for all $c \le 19$ and all (a, b) with $4ac - b^2 \le 1000$ (Appendix B). There are 3447 reduced forms with $0 < D \le 1000$, and $(-1)^{b+1} c_c^F(a, b) > 0$ for each of them.

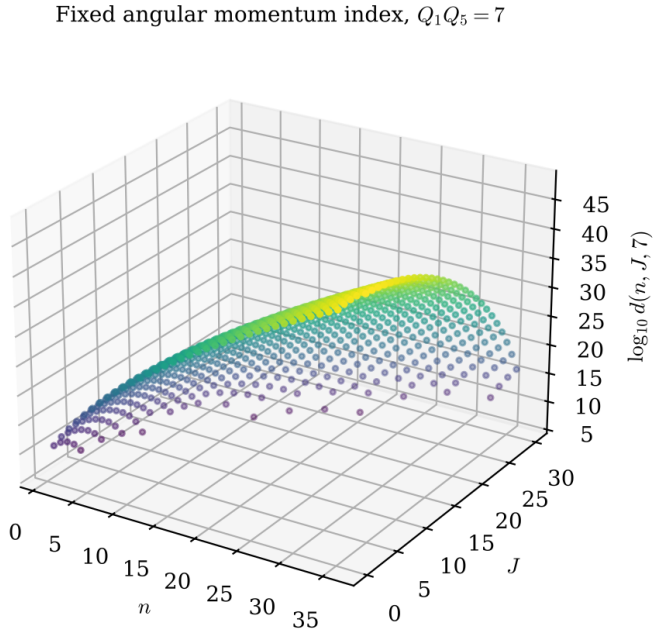


FIGURE 4. The index at fixed angular momentum $d(n, J, 7)$ for all charges with $Q_1Q_5 = 7$ and $0 < 4 \cdot 7\, n - J^2 \leq 1000$, on a logarithmic scale. For fixed n the index falls by many orders of magnitude from small J to the edge $J \approx 2\sqrt{7n}$ of the black hole region, following the discriminant $D = 28n - J^2$.

Five dimensions. Let $0 < D = 4mn - J^2 \leq 1000$. If $4(J+1) \geq D$, the second term of (6) vanishes and $d(n, J, m) = d^*(n, J, m) > 0$ by the four-dimensional part. Otherwise $J < D/4 \leq 250$, and m divides $(D + J^2)/4$; so there are finitely many such charges, 568 879 in all. For each of them we computed both terms of (6) exactly through their reduced forms, and $d(n, J, m) > 0$ in every case. □

The smallest relative margin $d(n, J, m)/d^*(n, J, m)$ among these charges is 0.18852, at $(m, n, J) = (13, 19, 0)$, where $D = 988$ and the second charge has $D' = 984$. Figure 5 shows all the margins.

11. REMARKS

The reduction cannot be dropped. If the index at fixed angular momentum is computed from the coefficients of a single function ψ_m^F, without passing to reduced forms, one obtains the raw difference $(-1)^{J+1}\big(c_m^F(n, J) - c_m^F(n, J+2)\big) = F_m(D) - F_m(D - 4(J+1))$, where F_m is extended to all $D \equiv -l^2 \pmod{4m}$ by Lemma 4.3 and by 0 when the coefficient lies below $n = -1$. Among the charges with $n \geq m$ and $D \leq 1000$ this raw difference is negative 16 times for $m = 5$ and 26 times for $m = 7$, and it is positive for $m = 1, 2, 3$. The failures occur because $F_m(D)$ is negative at small D, for instance $F_5(4) = -2\,023\,536$, or because the second charge is far from reduced. Figure 6 compares both prescriptions for $m = 5$. This is consistent with the attractor argument of Section 2: only the reduced form of a charge has its attractor point in the chamber where ψ_c^F counts single-centred states.

Composite and larger indices. For composite m two things fail. The value of $c_m^F(n, l)$ is no longer a function of D alone (Lemma 4.3), and $\Delta\psi_m^F$ is not a weak Jacobi form plus a multiple

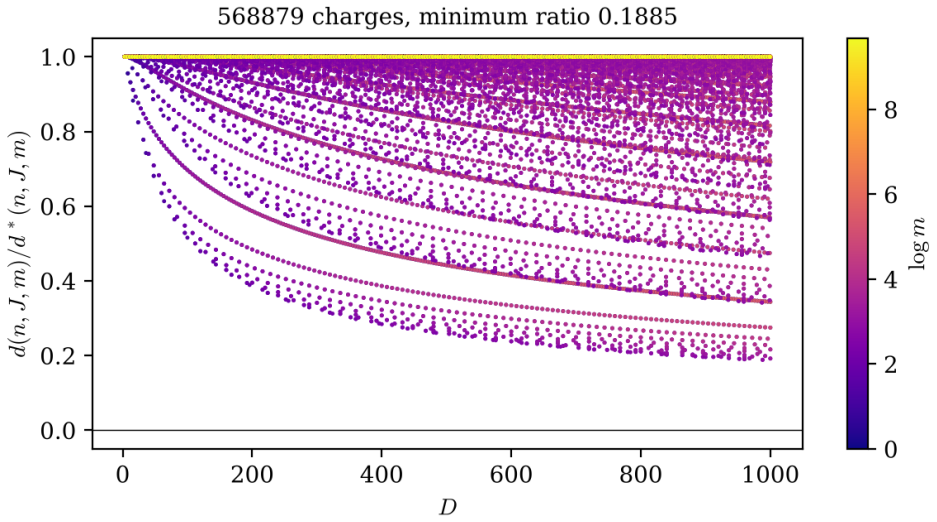


FIGURE 5. The relative margin $d(n, J, m)/d^*(n, J, m)$ for all five-dimensional charges with $D \le 1000$ and $4(J + 1) < D$, coloured by $\log m$. The margin is smallest when $J = 0$, where the two discriminants D and $D - 4$ are closest, and when m is large, where the growth rate $\alpha_m = \pi\sqrt{(m + 4)/m}$ is smallest.

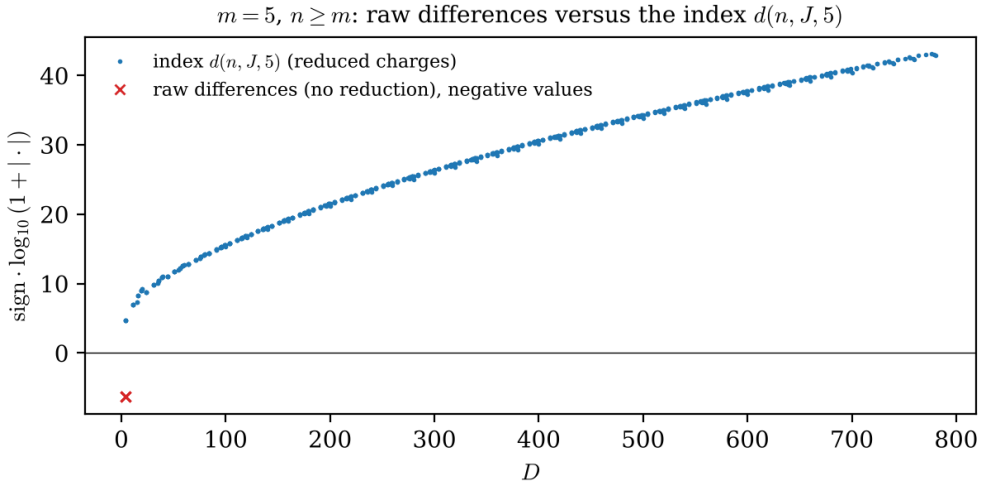


FIGURE 6. For $m = 5$ and $n \ge 5$: the index $d(n, J, 5)$, computed through reduced forms (dots, all positive), and the raw differences of coefficients of ψ_5^F where they are negative (crosses). The vertical axis is $\mathrm{sign}(x) \log_{10}(1 + |x|)$.

of $\mathcal{H}|V_m$ (Remark 5.5). Some other mock modular correction is then needed, and we have not determined it. For primes $m \ge 17$, Theorem D holds, but the certificate of Proposition 8.2, applied with the same choices, holds at $D_{17}^* = 1857$ and $D_{19}^* = 2375$ but fails one unit below, and these values lie beyond our tables, which end at $D \approx 1000$. Extending the tables to larger discriminants and repeating the finite part of Section 9 would extend Theorem B to these primes.

Size of the margins. The proof of Theorem B is tightest for $m = 1$, where $2\varepsilon_1(D_1^*)/\gamma_1(D_1^*) = 0.994$; there the error comes from the class number term, whose coefficient $\lambda_1 = 648$ is large compared with the main term at moderate D. For $m \geq 7$ the class number term is negligible and the next polar term, at $(D', l') = (-4m-(m-1)^2, m-1)$, limits the estimate. In the finite range the smallest relative margin is 0.1885 (Section 10); the index at fixed angular momentum is never close to zero relative to d^*.

Other charges. Theorems A and B concern charges of torsion one, for which the generating function $1/\Phi_{10}$ is known. For charges of higher torsion, and for the CHL orbifolds of type II string theory on K3, the method needs two inputs: an analogue of Theorem D that identifies the mock modular part of the finite part, and a tail estimate of the kind proved in Section 8.

Declarations

Data availability. The programs that compute the coefficients $c_m^F(n, l)$, verify the identities of Theorem D, evaluate the tail certificate in ball arithmetic and carry out the finite checks, together with the tables of coefficients they produce, are archived at `https://doi.org/10.5281/zenodo.0000000`.

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Ethics. Not applicable; the work involves no human or animal subjects.

Appendix A. Data

The weak Jacobi forms ϕ_m. Write $B = 2\phi_{0,1}$. The forms $\phi_m = \Delta\psi_m^F + \lambda_m \mathcal{H}|V_m$ of Theorem D are

$$\phi_1 = 3E_4 A, \qquad \phi_2 = \tfrac{11}{3} E_4 AB - \tfrac{10}{3} E_6 A^2,$$

and, for $m = 3, 5, 7, 11, 13$, the sums of the monomials below with the coefficients beside them.

$m = 3$:

E_4AB^2	$\frac{467}{192}$	$E_4^2A^3$	$\frac{203}{48}$
E_6A^2B	$-\frac{215}{48}$		

$m = 5$:

E_4AB^4	$\frac{72607}{165888}$	$E_4E_6A^4B$	$-\frac{1195}{324}$
$E_6A^2B^3$	$-\frac{16991}{10368}$	$E_6^2A^5$	$\frac{15041}{1296}$
$E_4^2A^3B^2$	$\frac{31561}{6912}$	$E_4^3A^5$	$-\frac{10813}{1152}$

$m = 7$:

E_4AB^6	$\frac{2522801}{63700992}$	$E_4^3A^5B^2$	$-\frac{1215121}{442368}$
$E_6A^2B^5$	$-\frac{397855}{1769472}$	$E_4^2E_6A^6B$	$-\frac{351565}{331776}$
$E_4^2A^3B^4$	$\frac{5487725}{5308416}$	$E_4E_6^2A^7$	$-\frac{1098143}{124416}$
$E_4E_6A^4B^3$	$-\frac{3377833}{1990656}$	$E_4^4A^7$	$\frac{356323}{36864}$
$E_6^2A^5B^2$	$\frac{317425}{55296}$		

$m = 11$:

E_4AB^{10}	$\frac{2131467191}{21134460321792}$	$E_6^3A^8B^3$	$\frac{540339337}{1289945088}$
$E_6A^2B^9$	$-\frac{5113428019}{52836150804 48}$	$E_4^3E_6A^8B^3$	$-\frac{6681275737}{20639121408}$
$E_4^2A^3B^8$	$\frac{13935457241}{17612050268 16}$	$E_4^2E_6^2A^9B^2$	$\frac{6195889207}{859963392}$
$E_4E_6A^4B^7$	$-\frac{970619809}{366917771392}$	$E_4^5A^9B^2$	$-\frac{59476197389}{9172942848}$
$E_6^2A^5B^6$	$\frac{5782066279}{41278242816}$	$E_4E_6^3A^{10}B$	$-\frac{1053477487}{80621568}$
$E_4^3A^5B^6$	$-\frac{22336678753}{660451885056}$	$E_4^4E_6A^{10}B$	$\frac{92843167663}{6879707136}$
$E_4^2E_6A^6B^5$	$-\frac{6566555011}{55037657088}$	$E_6^4A^{11}$	$\frac{404439233}{40310784}$
$E_4E_6^2A^7B^4$	$-\frac{459938771}{573308928}$	$E_4^3E_6^2A^{11}$	$-\frac{4042889417}{322486272}$
$E_4^4A^7B^4$	$\frac{68351027909}{55037657088}$	$E_4^6A^{11}$	$\frac{2092865525}{764411904}$

$m = 13$:

E_4AB^{12}	$\frac{21052966501}{60867245726 76096}$	$E_4^5A^9B^4$	$-\frac{168553028141}{293534171136}$
$E_6A^2B^{11}$	$-\frac{1691946635}{42268920643584}$	$E_4E_6^3A^{10}B^3$	$-\frac{37682583031}{46438023168}$
$E_4^2A^3B^{10}$	$\frac{101024148481}{25361352386 1504}$	$E_4^4E_6A^{10}B^3$	$\frac{62552439743}{55037657088}$
$E_4E_6A^4B^9$	$-\frac{39816293555}{23776267862016}$	$E_6^4A^{11}B^2$	$\frac{1064334965}{1289945088}$
$E_6^2A^5B^8$	$\frac{16000805975}{17612050268 16}$	$E_4^3E_6^2A^{11}B^2$	$-\frac{15342776135}{6879707136}$
$E_4^3A^5B^8$	$-\frac{7084900171}{14089640214528}$	$E_4^6A^{11}B^2$	$\frac{7920621283}{4076863488}$
$E_4^2E_6A^6B^7$	$-\frac{17239195301}{1320903770112}$	$E_4^2E_6^3A^{12}B$	$\frac{221053284 83}{7739670528}$
$E_4E_6^2A^7B^6$	$-\frac{25730368859}{990677827584}$	$E_4^5E_6A^{12}B$	$-\frac{38847685}{15925248}$
$E_4^4A^7B^6$	$\frac{81058056707}{8806025134 08}$	$E_4E_6^4A^{13}$	$-\frac{903349379}{362797056}$
$E_6^3A^8B^5$	$-\frac{507482165}{13759414272}$	$E_4^4E_6^2A^{13}$	$\frac{82753754885}{2293235712}$
$E_4^3E_6A^8B^5$	$\frac{3161174131}{55037657088}$	$E_4^7A^{13}$	$-\frac{1929616631}{2038431744}$
$E_4^2E_6^2A^9B^4$	$\frac{139668317891}{165112971264}$		

The constants are $\lambda_1 = 648$, $\lambda_2 = 3200$, $\lambda_3 = 25650$, $\lambda_5 = 1073720$, $\lambda_7 = 30178575$, $\lambda_{11} = 10914317934$ and $\lambda_{13} = 156883829400$; for $m \geq 2$ they are $p_{24}(m+1)$, and the linear algebra of Section 5 recovers each of them as the coefficient of $\mathcal{H}|V_m$.

Polar terms. The polar terms (D', l'), $0 \leq l' \leq m$, of ϕ_m/Δ and the coefficients $a_m(D', l')$ entering (19) are listed below. The first entry is the most polar term $(-m(m+4), m)$ with coefficient $m+2$. The entry $(-4m, 0)$ is the only one that differs from the corresponding coefficient of ψ_m^F, by $-\lambda_m\sigma(m)/12$.

$m = 1$:

$(-5,1)$ 3 $(-4,0)$ -6 $(-1,1)$ 816

$m = 2$:

$(-12,2)$ 4 $(-8,0)$ -152 $(-1,1)$ 11576
$(-9,1)$ 72 $(-4,2)$ 2752

$m = 3$:

$(-21,3)$ 5 $(-12,0)$ -2146 $(-1,1)$ 110826
$(-16,2)$ 96 $(-9,3)$ 10278
$(-13,1)$ 972 $(-4,2)$ 41250

$m = 5$:

$(-45,5)$ 7 $(-24,2)$ 12800 $(-9,3)$ 1304192
$(-36,4)$ 144 $(-21,1)$ 76955 $(-5,5)$ 2462695
$(-29,3)$ 1620 $(-20,0)$ -183052 $(-4,2)$ 2857380
$(-25,5)$ 2880 $(-16,4)$ 567964 $(-1,1)$ 6004640

$m = 7$:

$(-77, 7)$	9	$(-36, 6)$	46656	$(-16, 4)$	42700596
$(-64, 6)$	192	$(-32, 2)$	705030	$(-9, 3)$	42892335
$(-53, 5)$	2268	$(-29, 1)$	3222780	$(-8, 6)$	64504998
$(-49, 7)$	4032	$(-28, 0)$	−8155458	$(-4, 2)$	113142510
$(-44, 4)$	19200	$(-25, 5)$	10443525	$(-1, 1)$	213270204
$(-37, 3)$	128250	$(-21, 7)$	30703455		

$m = 11$:

$(-165, 11)$	13	$(-69, 5)$	7516040	$(-36, 6)$	11068933758
$(-144, 10)$	288	$(-64, 8)$	4924800	$(-33, 11)$	10985997934
$(-125, 9)$	3564	$(-60, 4)$	35582984	$(-25, 5)$	11626016334
$(-121, 11)$	6336	$(-56, 10)$	8294400	$(-20, 8)$	22171278828
$(-108, 8)$	32000	$(-53, 3)$	150895143	$(-16, 4)$	24728241564
$(-100, 10)$	77760	$(-49, 7)$	29611008	$(-12, 10)$	33235440206
$(-93, 7)$	230850	$(-48, 2)$	572889900	$(-9, 3)$	32215422180
$(-81, 9)$	691200	$(-45, 1)$	1923116507	$(-5, 7)$	45397013391
$(-80, 6)$	1410048	$(-44, 0)$	−5383354674	$(-4, 2)$	54013959228
$(-77, 11)$	944784	$(-37, 9)$	10972492134	$(-1, 1)$	95512792158

$m = 13$:

$(-221, 13)$	15	$(-88, 6)$	47443968	$(-40, 12)$	209835079200
$(-196, 12)$	336	$(-81, 9)$	38071296	$(-36, 6)$	239671505556
$(-173, 11)$	4212	$(-77, 5)$	211250034	$(-29, 9)$	473086686780
$(-169, 13)$	7488	$(-69, 11)$	74795400	$(-25, 5)$	226363597200
$(-152, 10)$	38400	$(-68, 4)$	859106592	$(-17, 11)$	317715802800
$(-144, 12)$	93312	$(-65, 13)$	92160000	$(-16, 4)$	453680614524
$(-133, 9)$	282150	$(-64, 8)$	206154240	$(-13, 13)$	632140801722
$(-121, 11)$	844800	$(-61, 3)$	3196426050	$(-12, 8)$	325296962928
$(-117, 13)$	1154736	$(-56, 2)$	10826901840	$(-9, 3)$	353837671920
$(-116, 8)$	1762560	$(-53, 1)$	32893848945	$(-4, 2)$	1079018682192
$(-101, 7)$	9663480	$(-52, 0)$	−96093457164	$(-1, 1)$	1672865232240
$(-100, 10)$	6156000	$(-49, 7)$	53290933296		
$(-92, 12)$	10368000	$(-48, 10)$	78898770252		

Appendix B. The computations

All coefficients were computed in exact integer or rational arithmetic with the FLINT library through its Python interface; the numerical bounds of Section 8 were evaluated in Arb ball arithmetic [17]. The programs and the resulting tables are in the archive cited in the Declarations. The steps are as follows.

The form $2\phi_{0,1}$. We used $\phi_{0,1}(\tau, z) = 4\sum_{i=2}^{4} \vartheta_i(\tau, z)^2/\vartheta_i(\tau, 0)^2$. The right side is a weak Jacobi form of weight 0 and index 1: each quotient is entire in z and has an expansion in nonnegative powers of $q^{1/2}$, the half-integral powers cancel between the quotients for $i = 3$ and $i = 4$, and under $\tau \mapsto \tau + 1$ and $(\tau, z) \mapsto (-1/\tau, z/\tau)$ the three quotients are permuted, the second transformation multiplying each of them by $\mathbf{e}(z^2/\tau)$. The only such form up to scalars is $\phi_{0,1}$ by (8), and the value 12 at $z = 0$ fixes the scalar. Only the coefficients of ζ^0 and ζ^1 are needed, since $C(4n-l^2)$ depends on $4n-l^2$ alone; we computed $C(D)$ for $D \le 4(m+1)(K+1)$, which covers every argument that occurs below.

The functions χ_N. Applying $p\,\frac{d}{dp}$ to (1) gives $N\chi_N = \sum_{k=1}^{N} kL_k\,\chi_{N-k}$ with $L_k = (2\phi_{0,1})|V_k$, and by (11) the coefficient of $q^n\zeta^l$ in kL_k is $\sum_{r|(k,n,l)}(k/r)\,C\big((4nk - l^2)/r^2\big)$. All series are truncated at q^K and stored as polynomials in ζ and ζ^{-1}.

The functions ψ_m^F. The product $\Delta\psi_m = \chi_{m+1}\cdot A^{-1}$ was computed with the expansion of $1/A$ in the strip $|q|<|\zeta|<1$ from Section 4, the expansion of $A_{1,m}$ in the same strip was subtracted with the factor $p_{24}(m+1)$, and the result was divided by Δ by multiplying with $\sum_k p_{24}(k)q^{k-1}$. This gives $c_m^F(n,l)$ for $0\le l\le m$ and $-1\le n<K$. We did this for $1\le m\le 19$ with $K=\lfloor(1000+m^2)/4m\rfloor+2$, which covers every reduced form of discriminant at most 1000. By Lemma 4.2 these coefficients determine all others.

Theorem D. The coefficients of $q^n\zeta^l$, $0\le n\le 14$, $0\le l\le m$, of $\Delta\psi_m^F$, of the monomials $E_4^\alpha E_6^\beta A^i B^{m-i}$ and of $\mathcal{H}|V_m$ give an overdetermined linear system over $\mathbb{Q}$, solved exactly. Its matrix has full column rank for every $m\in\mathcal{M}$, and the solution is the one in Appendix A.

Reduced forms and Theorem A. Reduced forms were computed by the algorithm recalled after Lemma 3.2, and each output was checked to satisfy $0\le b\le c\le a$ and $4ac-b^2=D$.

The exact formula. As a test of Theorem 6.1, the terms with $c=1$ of (19) were evaluated with 50-digit arithmetic at the largest n available for each m and compared with the exact coefficients $a_m(n,l)=c_m^F(n,l)+\lambda_m\delta_m(n,l)$. The relative differences are below $2\cdot10^{-25}$ for $m=13$ and smaller for smaller m (for instance $2.5\cdot10^{-49}$ for $m=1$), consistent with the size of the terms with $c=2$.

The tail certificate. The quantities of Proposition 8.2 were evaluated in Arb with a working precision of 200 bits.

Appendix C. Two elementary bounds

Lemma C.1. *For $N\ge1$, $0\le H(N)\le N+1$; and $|H(0)|=\frac{1}{12}$.*

Proof. $H(N)$ is a weighted count, with weights at most one, of the $\mathrm{SL}_2(\mathbb{Z})$-classes of positive definite forms of discriminant $-N$. Each class contains a form $[a,b,c]$ with $|b|\le a\le c$ [9, §2.2], and then $N=4ac-b^2\ge3a^2$, so $a\le t=\sqrt{N/3}$; moreover $c=(N+b^2)/4a$ is determined by a and b. The number of pairs (a,b) with $1\le a\le t$ and $|b|\le a$ is at most $\sum_{a\le t}(2a+1)\le t^2+2t\le\frac{2N}{3}+1$, using $2t\le t^2+1$. □

Lemma C.2. *For $k\ge2$, $p_{24}(k)\le c_0\,(k-1)^{-6}e^{4\pi\sqrt{k-1}}$ with $c_0=\prod_{n\ge1}(1-e^{-2\pi n})^{-24}$.*

Proof. The coefficients of $\Delta^{-1}=\sum_k p_{24}(k)q^{k-1}$ are positive, so for $t>0$ and $q=e^{-2\pi t}$ one has $p_{24}(k)e^{-2\pi(k-1)t}\le\Delta(it)^{-1}$. The modularity $\Delta(-1/\tau)=\tau^{12}\Delta(\tau)$ at $\tau=it$ gives $\Delta(it)^{-1}=t^{12}\Delta(i/t)^{-1}=t^{12}e^{2\pi/t}\prod_{n\ge1}(1-e^{-2\pi n/t})^{-24}$. Take $t=(k-1)^{-1/2}\le1$; then the product is at most c_0, and $p_{24}(k)\le e^{2\pi\sqrt{k-1}}(k-1)^{-6}e^{2\pi\sqrt{k-1}}c_0$. □

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